

# Business Intelligence Capstone

Data Analysis, Visualisation and Insights

Final Project Presentation

Prepared using Excel, PostgreSQL and Tableau



- 
- Introduction
  - Methodology
  - Results
  - Discussion
  - Conclusion
  - Appendix



## Introduction: Business Scenario

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- The analysis is based on a retail chain operating across multiple stores, cities and product categories.
- The business produces sales, revenue, stock, price, store, city and product data that can be used to understand performance.
- The raw data is distributed across separate files/tables and therefore requires cleaning, preparation and consolidation before analysis.
- The business objective is to turn operational data into evidence for better sales, inventory and product decisions.

## Introduction: Purpose of the Analysis

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- Identify sales and revenue trends across time periods, stores and regions.
- Compare city-level and product-level performance to identify stronger and weaker areas of the business.
- Explore whether stock availability is associated with sales volume.
- Use regression analysis to assess the relationship between sales and selected independent variables such as date, stock and price.
- Communicate findings through Excel charts, SQL outputs and Tableau dashboards.



# Methodology

Data preparation, querying, statistical analysis and visualisation workflow

## Data Details

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- Started with multiple retail datasets containing sales, store, city, product and product hierarchy information.
- The main sales dataset included fields such as sales date, store ID, product ID, sales volume, revenue, stock levels and price.
- Supporting datasets were used to enrich the sales data with store names, city names, product names and product hierarchy details.
- The data was cleaned and combined to enable analysis by date, store, city, product and product category.

### Why this matters

The data could not answer business questions effectively while key information was split across separate files. Consolidation created an analysis-ready view of the retail operation.

## Module 1 Methodology: Excel Cleaning and Pivot Tables

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- Imported the source files into Excel and Power Query.
- Checked and corrected formatting issues, unclear column names, inconsistent text values, duplicates and missing values where relevant.
- Joined related datasets using keys such as store ID, city ID and product ID.
- Created pivot tables to summarise total sales, total revenue and average sales by city, store and product.

## Module 2 Methodology: PostgreSQL Querying and Analysis

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- Loaded the prepared retail data into the BICapstone PostgreSQL database.
- Validated table creation and data population in pgAdmin.
- Used SQL joins to combine sales data with store, city and product information.
- Applied grouping, aggregation, ROLLUP and time formatting to calculate sales performance, hierarchy summaries and trends across regions.

## Module 3 Methodology: Excel Visualisation and Statistics

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- Created Excel charts to show trends, comparisons, hierarchical product performance and stock-sales relationships.
- Used a line chart for time-based performance, a bar chart for category comparisons, a sunburst chart for hierarchy and a scatter plot for correlation exploration.
- Used Excel Analysis ToolPak to run regression analysis.
- Interpreted the regression output using R-squared, coefficients and p-values.

## Module 4 Methodology: Tableau Visualisations and Dashboard

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- Connected Tableau to the prepared retail dataset.
- Created basic visualisations: line chart, bar chart, scatter plot and packed bubbles chart.
- Used Tableau concepts such as Measure Names, Measure Values and dashboard actions to support multi-measure analysis and interactivity.
- Built a dashboard to communicate sales, product, regional and stock-related insights in one view.



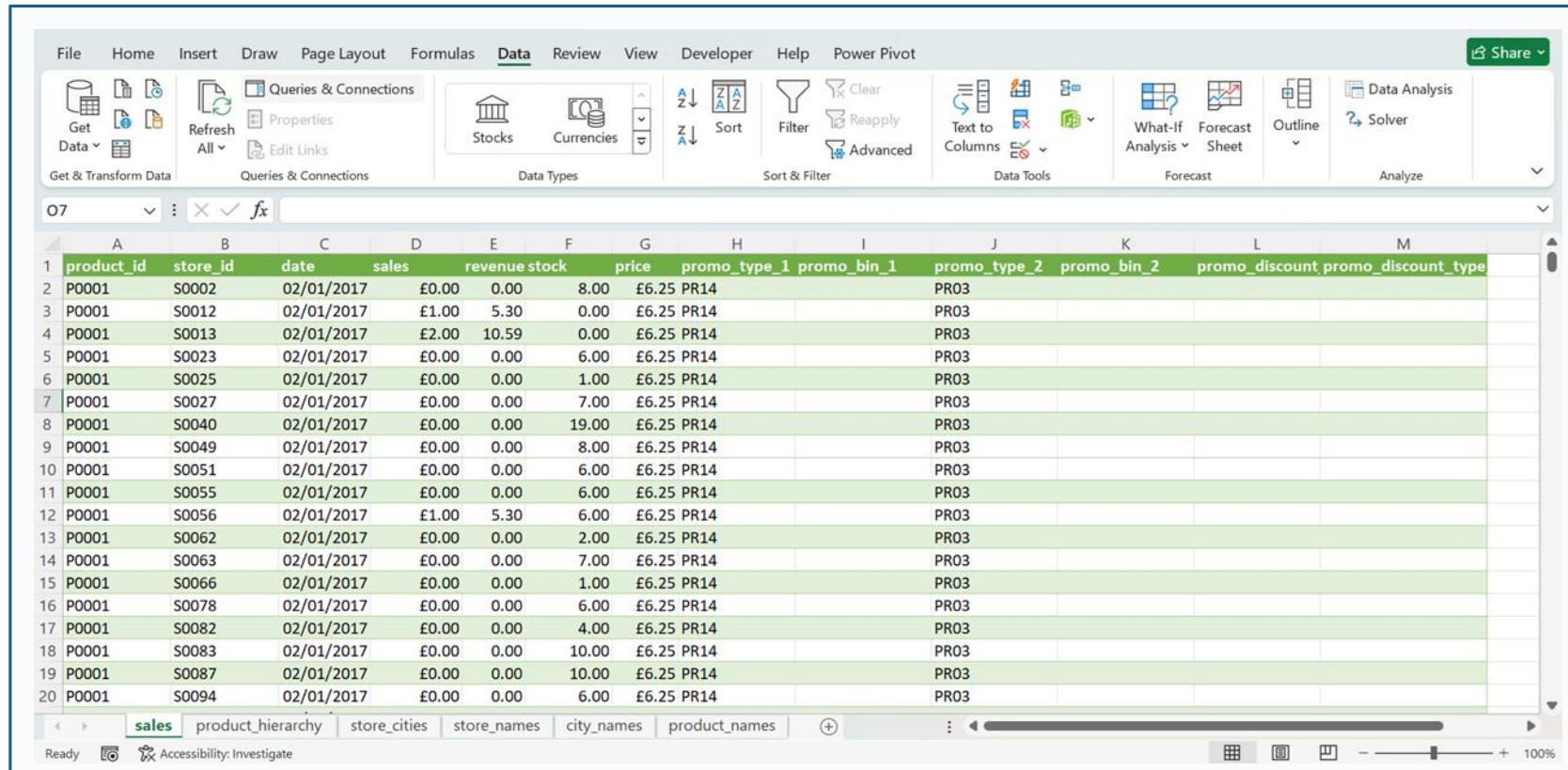
# Results

## Results Overview

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- The results section follows the required module and lesson structure.
- Each task slide includes a screenshot for the completed output and short interpretation text.
- The evidence covers cleaned data, pivot tables, pgAdmin, SQL queries, SQL outputs, Excel charts, regression analysis, Tableau visualisations and Tableau dashboard interactivity.

# Module 1 Lesson 1: Cleaned and Prepared Dataset



|    | A          | B        | C          | D     | E       | F     | G     | H            | I           | J            | K           | L              | M                   |
|----|------------|----------|------------|-------|---------|-------|-------|--------------|-------------|--------------|-------------|----------------|---------------------|
|    | product_id | store_id | date       | sales | revenue | stock | price | promo_type_1 | promo_bin_1 | promo_type_2 | promo_bin_2 | promo_discount | promo_discount_type |
| 1  | P0001      | S0002    | 02/01/2017 | £0.00 | 0.00    | 8.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 2  | P0001      | S0012    | 02/01/2017 | £1.00 | 5.30    | 0.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 3  | P0001      | S0013    | 02/01/2017 | £2.00 | 10.59   | 0.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 4  | P0001      | S0023    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 5  | P0001      | S0025    | 02/01/2017 | £0.00 | 0.00    | 1.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 6  | P0001      | S0027    | 02/01/2017 | £0.00 | 0.00    | 7.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 7  | P0001      | S0040    | 02/01/2017 | £0.00 | 0.00    | 19.00 | £6.25 | PR14         |             | PR03         |             |                |                     |
| 8  | P0001      | S0049    | 02/01/2017 | £0.00 | 0.00    | 8.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 9  | P0001      | S0051    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 10 | P0001      | S0055    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 11 | P0001      | S0056    | 02/01/2017 | £1.00 | 5.30    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 12 | P0001      | S0062    | 02/01/2017 | £0.00 | 0.00    | 2.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 13 | P0001      | S0063    | 02/01/2017 | £0.00 | 0.00    | 7.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 14 | P0001      | S0066    | 02/01/2017 | £0.00 | 0.00    | 1.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 15 | P0001      | S0078    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 16 | P0001      | S0082    | 02/01/2017 | £0.00 | 0.00    | 4.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 17 | P0001      | S0083    | 02/01/2017 | £0.00 | 0.00    | 10.00 | £6.25 | PR14         |             | PR03         |             |                |                     |
| 18 | P0001      | S0087    | 02/01/2017 | £0.00 | 0.00    | 10.00 | £6.25 | PR14         |             | PR03         |             |                |                     |
| 19 | P0001      | S0094    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |
| 20 | P0001      | S0094    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03         |             |                |                     |

- The cleaned dataset provided the foundation for all later analysis.
- Separate sales, store, city and product files were made consistent and prepared for analysis.
- Cleaning reduced the risk of unreliable outputs caused by inconsistent columns, formatting issues or incomplete data.
- The prepared dataset enabled accurate pivot tables, SQL queries, charts, regression analysis and Tableau dashboards.

## Module 1 Lesson 2: Pivot Table — Sales Data Analysis

The screenshot displays an Excel spreadsheet with a PivotTable summarizing sales data by store. The PivotTable is located in the range A1:E11. The columns are labeled 'Store\_names', 'Total Sales', 'Total Revenue', 'Total Stock', and 'Average Sales'. The rows list various stores, including AGD Centrum, AGD Expert, Appliance Centre, Appliance Direct, Appliance Paradise, Appliance World, Bimeks, Block, Boulanger (National Chain), Casa da Tecnologia, Casa Digital, Casa Tech, Cecchini, Centro delloElettronica, Comet, Conrad Electronic (National Chain), Coolblue, Cosmos Sport, and Currys.

On the right side of the spreadsheet, the 'PivotTable Fields' task pane is open. It shows the following configuration:

- Choose fields to add to report:** A search bar and a list of fields. The fields 'sales', 'revenue', and 'stock' are checked.
- Columns:** A dropdown menu showing 'Σ Values'.
- Rows:** A dropdown menu showing 'store\_name'.
- Σ Values:** A dropdown menu showing 'Total Sales'.
- Defer Layout:** A checkbox that is unchecked.
- Update:** A button to refresh the data.

The status bar at the bottom of the spreadsheet shows 'Ready' and 'Accessibility: Investigate'.

- Summarised key measures such as total sales, total revenue, average sales and stock levels.
- This helped compare sales performance across the chosen business dimension.
- Business value: managers can use this output to monitor performance and prioritise operational decisions.

## Module 1 Lesson 2: Pivot Table — Sales by City

The screenshot shows an Excel spreadsheet with a PivotTable titled "Sales by Cities". The PivotTable is located in the range A2:E20. The columns are City Name, Store Count, Total Sales, Total Revenue, and Average Sales. The data lists 20 cities with their respective sales metrics. The PivotTable Fields task pane on the right shows the fields included in the report: store\_id, sales, and revenue.

|    | A                      | B           | C           | D             | E             |
|----|------------------------|-------------|-------------|---------------|---------------|
| 1  | <b>Sales by Cities</b> |             |             |               |               |
| 2  | City Name              | Store Count | Total Sales | Total Revenue | Average Sales |
| 3  | Barcelona              | 433         | 109         | £209.53       | £0.25         |
| 4  | Belgrade               | 1601        | 599         | £1,409.15     | £0.37         |
| 5  | Berlin                 | 786         | 505         | £1,119.53     | £0.64         |
| 6  | Brussels               | 882         | 337         | £652.34       | £0.38         |
| 7  | Budapest               | 936         | 849         | £1,673.98     | £0.91         |
| 8  | Copenhagen             | 778         | 275         | £596.66       | £0.35         |
| 9  | Dublin                 | 1638        | 408         | £843.64       | £0.25         |
| 10 | Edinburgh              | 697         | 335         | £875.68       | £0.48         |
| 11 | Florence               | 1652        | 469         | £1,381.19     | £0.28         |
| 12 | Frankfurt              | 503         | 72          | £197.89       | £0.14         |
| 13 | Geneva                 | 494         | 83          | £197.59       | £0.17         |
| 14 | Helsinki               | 3793        | 2387        | £5,273.97     | £0.63         |
| 15 | Istanbul               | 661         | 326         | £796.99       | £0.49         |
| 16 | Krakow                 | 514         | 165         | £427.57       | £0.32         |
| 17 | Kyiv                   | 397         | 152         | £364.00       | £0.38         |
| 18 | Lisbon                 | 620         | 198         | £494.08       | £0.32         |
| 19 | London                 | 12482       | 9505        | £32,143.98    | £0.76         |
| 20 | Luxembourg             | 1219        | 204         | £683.62       | £0.17         |

- Compared sales and revenue performance across geographical locations.
- Provided evidence for regional marketing, stock allocation and resource planning.
- Business value: city-level performance can guide location-specific sales strategies.

## Module 1 Lesson 2: Pivot Table — Product Performance

The screenshot shows a Google Sheet with a Pivot Table titled "Sales by Products". The Pivot Table is located in the range A2:E20. The columns are Product Name, Total Sales, Total Revenue, and Average Sales. The rows list 20 different products. To the right of the Pivot Table is the "PivotTable Fields" sidebar, which allows users to choose fields to add to the report. The fields listed are product\_id, product\_length, product\_depth, product\_width, cluster\_id, hierarchy1\_id, hierarchy2\_id, hierarchy3\_id, hierarchy4\_id, hierarchy5\_id, product\_name, sales.1, and revenue. The fields product\_name, sales.1, and revenue are checked, indicating they are included in the report.

|    | A                                   | B                  | C                    | D                    | E | F |
|----|-------------------------------------|--------------------|----------------------|----------------------|---|---|
| 1  | <b>Sales by Products</b>            |                    |                      |                      |   |   |
| 2  | <b>Product Name</b>                 | <b>Total Sales</b> | <b>Total Revenue</b> | <b>Average Sales</b> |   |   |
| 3  | NanoGrillflow                       | 4268               | £1,013.12            | 9.38                 |   |   |
| 4  | Handheld Milk Frother               | 1786               | £4,444.85            | 4.15                 |   |   |
| 5  | CyberHeaterhub                      | 1346               | £616.53              | 3.29                 |   |   |
| 6  | Smart Speakers (with voice assistar | 1232               | £798.64              | 2.80                 |   |   |
| 7  | MaxDryerwave                        | 1084               | £5,128.99            | 2.79                 |   |   |
| 8  | TechTVlux                           | 849                | £795.26              | 2.02                 |   |   |
| 9  | SolarBlenderlux                     | 712                | £2,699.16            | 4.32                 |   |   |
| 10 | EcoVacuumcast                       | 547                | £949.55              | 1.24                 |   |   |
| 11 | SmartFridgewave                     | 509                | £1,247.25            | 1.15                 |   |   |
| 12 | Pizza Oven                          | 452                | £482.11              | 1.48                 |   |   |
| 13 | TechBlenderpulse                    | 437                | £740.62              | 10.93                |   |   |
| 14 | SmartBlenderflow                    | 428                | £697.57              | 1.07                 |   |   |
| 15 | SmartFridgedrive                    | 416                | £293.79              | 0.92                 |   |   |
| 16 | Portable Heater                     | 406                | £5,567.42            | 0.95                 |   |   |
| 17 | SmartVacuumcast                     | 370                | £428.16              | 1.12                 |   |   |
| 18 | SolarGrillpulse                     | 330                | £1,462.33            | 0.76                 |   |   |
| 19 | PowerDryerflow                      | 328                | £360.21              | 1.30                 |   |   |
| 20 | CyberMixermatic                     | 320                | £221.94              | 0.82                 |   |   |

- Compared product performance using sales, revenue and average sales measures.
- Supported product strategy, promotion planning, pricing review and inventory control.
- Business value: product-level evidence helps the business focus on items that contribute most to revenue.

# Module 1 Results: Strategic Significance

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## Sales data analysis

Shows overall performance and highlights where the retail chain is generating stronger or weaker results.

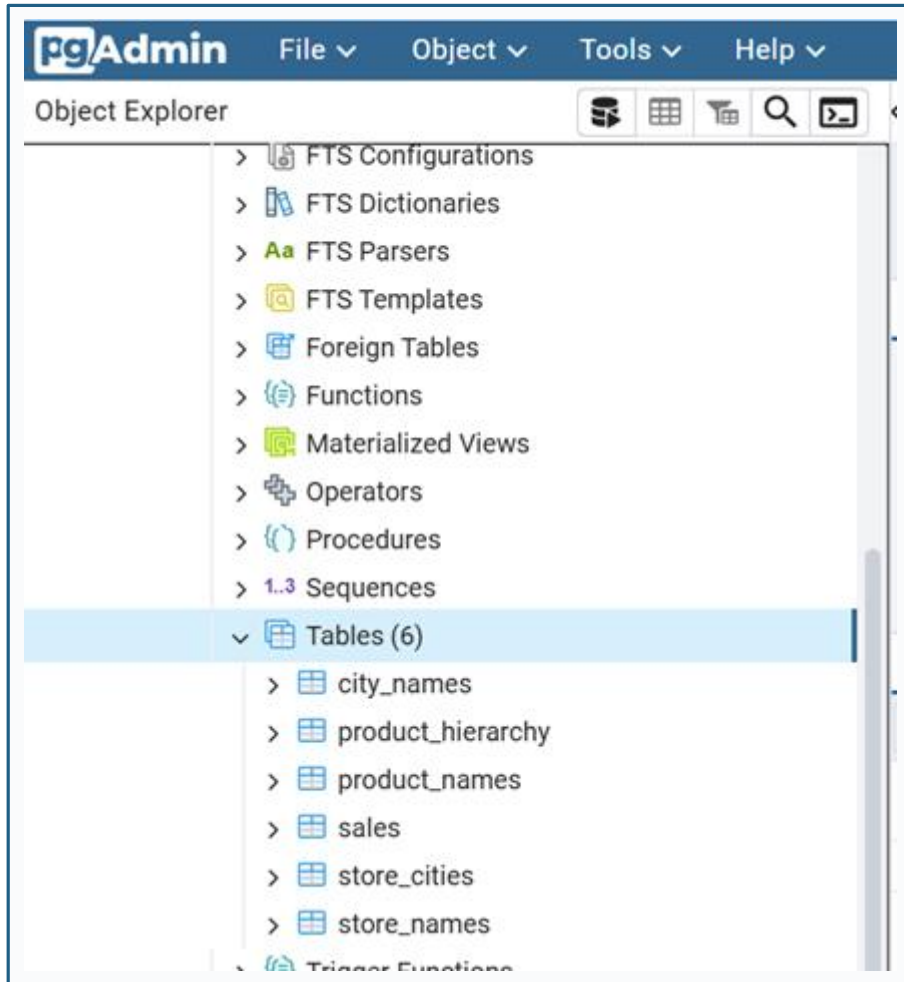
## Sales by city

Supports regional decisions, including local marketing, staffing, store-level planning and stock distribution.

## Product performance

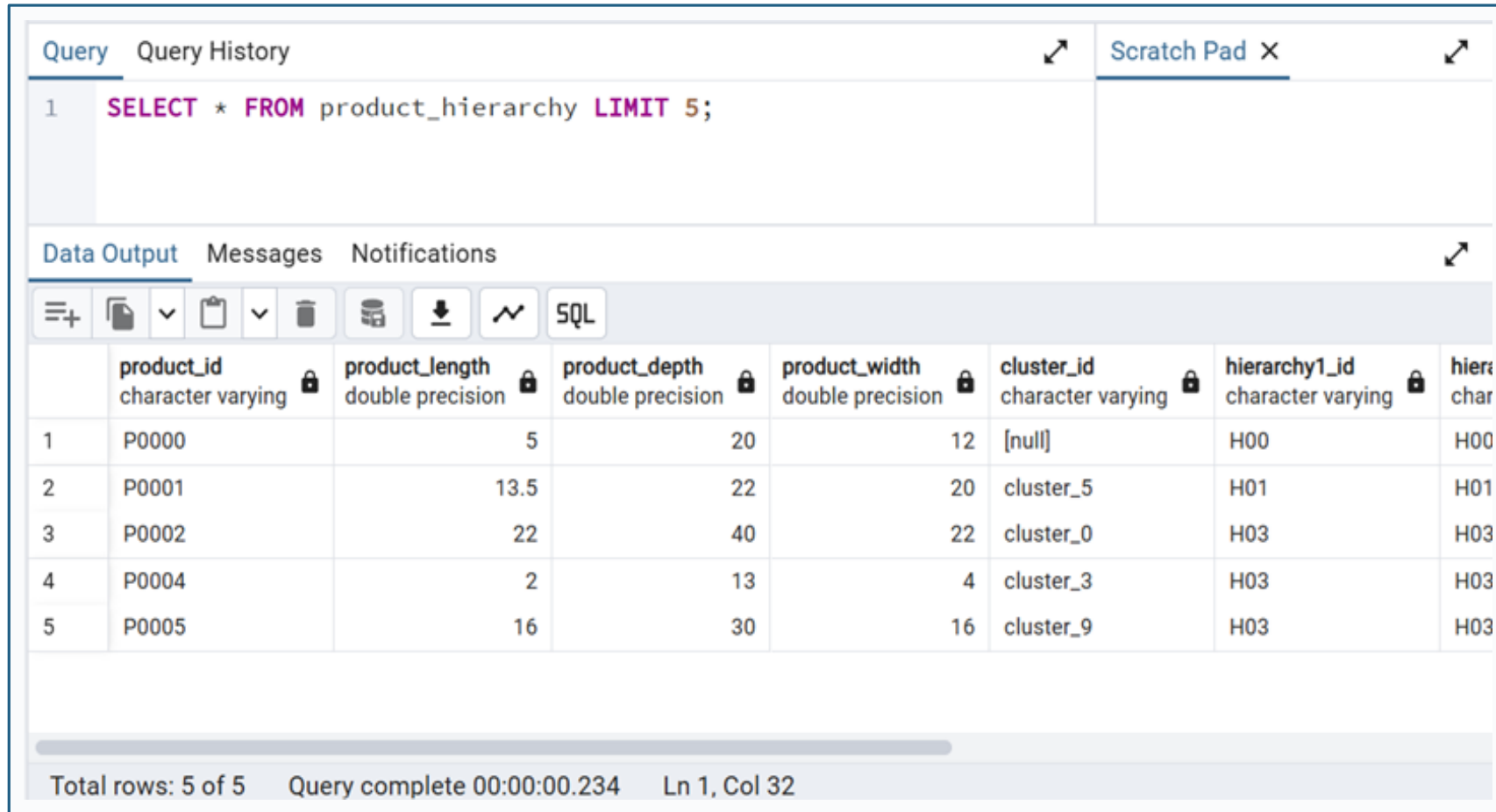
Shows which products and categories drive sales and which may need promotion, pricing review or stock adjustment.

## Module 2 Lesson 1: pgAdmin Interface and BICapstone Tables



- This pgAdmin interface confirms that the PostgreSQL database and tables were created successfully.
- The database structure made the retail datasets available for repeatable SQL querying.
- This step established the technical environment for later analysis.

## Module 2 Lesson 1: Data Population Check



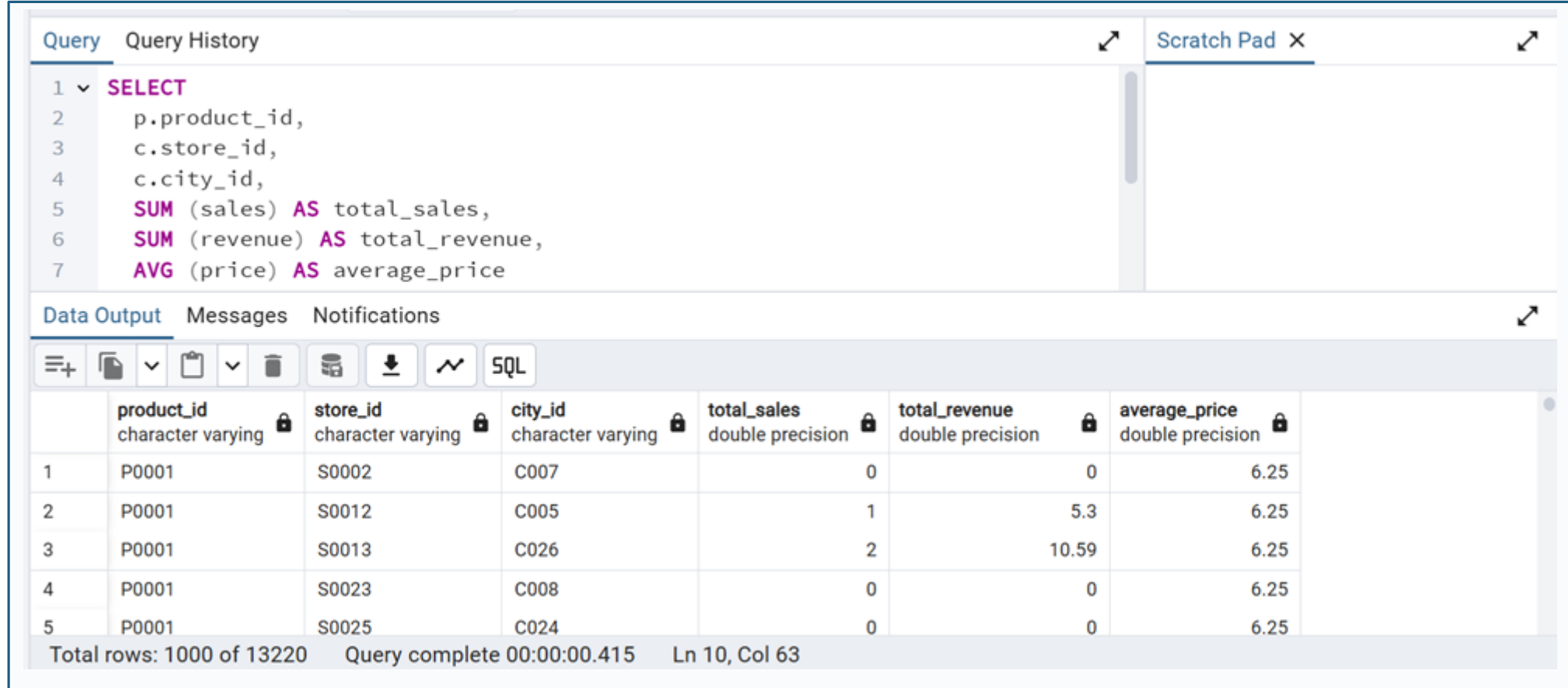
The screenshot shows a database query interface. At the top, there are tabs for 'Query' and 'Query History'. The 'Query' tab is active, displaying a SQL query: `SELECT * FROM product_hierarchy LIMIT 5;`. To the right of the query is a 'Scratch Pad' tab. Below the query, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table of results. The table has 8 columns: `product_id` (character varying), `product_length` (double precision), `product_depth` (double precision), `product_width` (double precision), `cluster_id` (character varying), `hierarchy1_id` (character varying), and `hierarch` (character). The results show 5 rows of data. At the bottom, a status bar indicates 'Total rows: 5 of 5', 'Query complete 00:00:00.234', and 'Ln 1, Col 32'.

|   | product_id<br>character varying | product_length<br>double precision | product_depth<br>double precision | product_width<br>double precision | cluster_id<br>character varying | hierarchy1_id<br>character varying | hierarch<br>character |
|---|---------------------------------|------------------------------------|-----------------------------------|-----------------------------------|---------------------------------|------------------------------------|-----------------------|
| 1 | P0000                           | 5                                  | 20                                | 12                                | [null]                          | H00                                | H00                   |
| 2 | P0001                           | 13.5                               | 22                                | 20                                | cluster_5                       | H01                                | H01                   |
| 3 | P0002                           | 22                                 | 40                                | 22                                | cluster_0                       | H03                                | H03                   |
| 4 | P0004                           | 2                                  | 13                                | 4                                 | cluster_3                       | H03                                | H03                   |
| 5 | P0005                           | 16                                 | 30                                | 16                                | cluster_9                       | H03                                | H03                   |

Total rows: 5 of 5    Query complete 00:00:00.234    Ln 1, Col 32

- This query checked whether data was successfully loaded into one of the tables.
- Validating table contents before analysis prevents conclusions being drawn from missing or incorrectly imported data.

## Module 2 Lesson 1: SQL Sales Performance Analysis



The screenshot displays a SQL IDE interface. The top section shows a query editor with a SQL query. The bottom section shows the results of the query in a table format. The query is as follows:

```
1 SELECT
2   p.product_id,
3   c.store_id,
4   c.city_id,
5   SUM (sales) AS total_sales,
6   SUM (revenue) AS total_revenue,
7   AVG (price) AS average_price
```

The results table has the following columns: product\_id, store\_id, city\_id, total\_sales, total\_revenue, and average\_price. The data is as follows:

|   | product_id | store_id | city_id | total_sales | total_revenue | average_price |
|---|------------|----------|---------|-------------|---------------|---------------|
| 1 | P0001      | S0002    | C007    | 0           | 0             | 6.25          |
| 2 | P0001      | S0012    | C005    | 1           | 5.3           | 6.25          |
| 3 | P0001      | S0013    | C026    | 2           | 10.59         | 6.25          |
| 4 | P0001      | S0023    | C008    | 0           | 0             | 6.25          |
| 5 | P0001      | S0025    | C024    | 0           | 0             | 6.25          |

Total rows: 1000 of 13220 Query complete 00:00:00.415 Ln 10, Col 63

- The query summarised sales, revenue and stock by store.
- It helped identify stronger and weaker stores for performance review.

## Module 2 Lesson 1: Value of Database Setup

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- The database organised the data into structured relational tables.
- Primary business dimensions such as stores, cities and products could be joined to the sales data.
- SQL enabled consistent filtering, aggregation and repeatable analysis.
- Using PostgreSQL made it easier to scale from simple checks to multidimensional summaries and trend analysis.
- The database therefore acted as the analytical foundation for subsequent SQL-based business intelligence tasks.

## Module 2 Lesson 2: Data Cube with ROLLUP

The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT
2     store_id,
3     TO_CHAR("date", 'YYYY-MM') AS sales_month,
4     SUM(sales) AS total_sales
5 FROM
6     sales
7 GROUP BY
8     ROLLUP (
9         store_id,
10        TO_CHAR("date", 'YYYY-MM')
11    )
```

The results pane shows the output of the query. It includes a toolbar with icons for expand, save, undo, redo, delete, refresh, download, and a graph. The results are displayed in a table with three columns: store\_id, sales\_month, and total\_sales. The first row shows a grand total for all data, and the second row shows a subtotal for a specific store.

|   | store_id<br>character varying | sales_month<br>text | total_sales<br>double precision |
|---|-------------------------------|---------------------|---------------------------------|
| 1 | [null]                        | [null]              | 9856.880999999998               |
| 2 | S0013                         | 2017-03             | 27                              |

Total rows: 352 of 352    Query complete 00:00:00.227    Ln 11, Col 7

- ROLLUP produced subtotals and grand totals across selected dimensions.
- This allowed detailed results and summary-level results to be reviewed together.

## Module 2 Lesson 2: Summarising Data Along Hierarchies

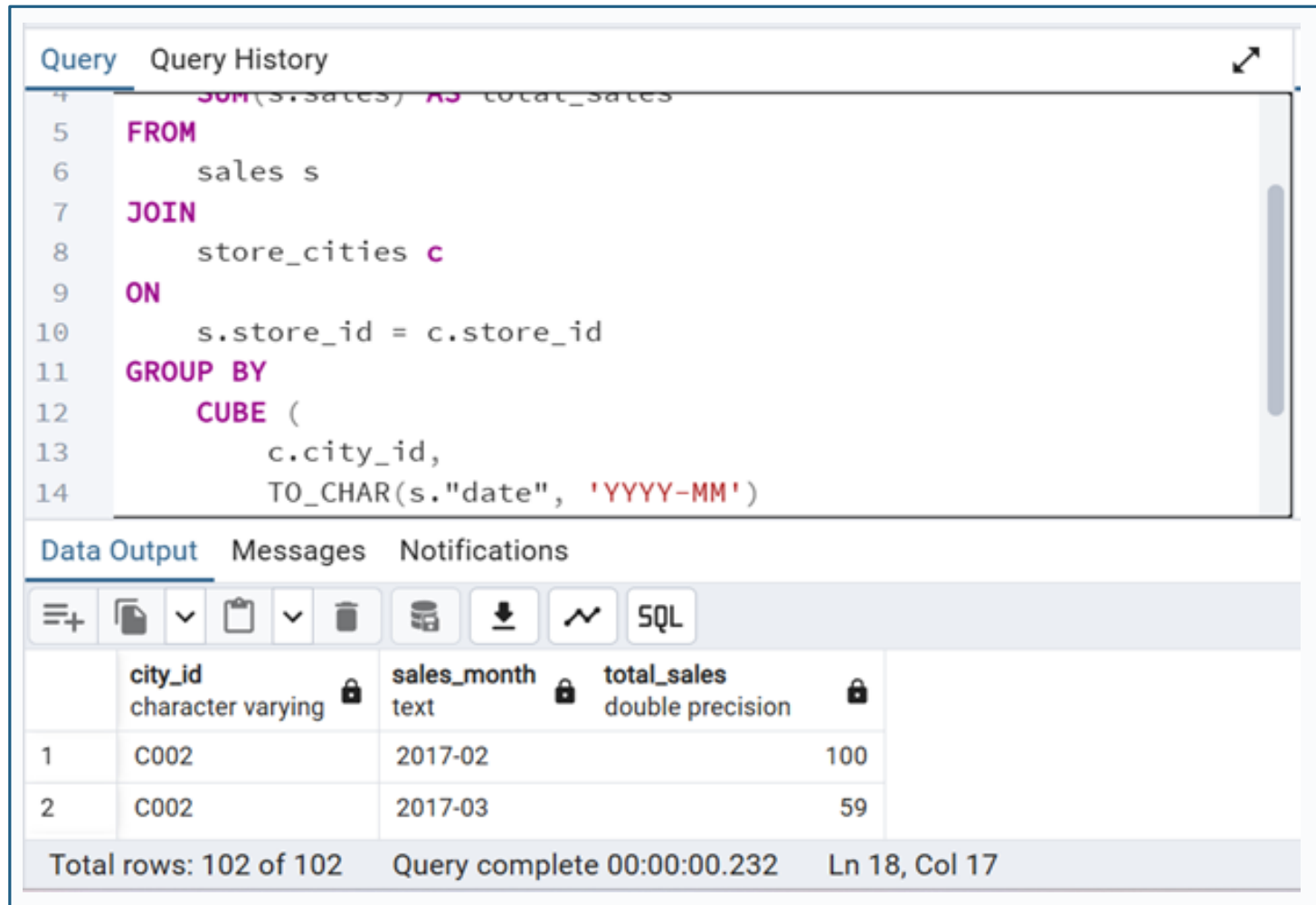
| Query |                               | Query History |  |
|-------|-------------------------------|---------------|--|
| 1     | SELECT                        |               |  |
| 2     | hierarchy1_id, hierarchy2_id, |               |  |
| 3     | SUM(s.sales) AS total_sales   |               |  |
| 4     | FROM                          |               |  |
| 5     | sales s                       |               |  |
| 6     | JOIN                          |               |  |
| 7     | product_hierarchy p           |               |  |

| Data Output                                                          |                                    | Messages                           |                                 | Notifications |    |
|----------------------------------------------------------------------|------------------------------------|------------------------------------|---------------------------------|---------------|----|
| ≡+                                                                   | 📄                                  | ▼                                  | 📋                               | ▼             | 🗑️ |
| 🗄️                                                                   | ⬇️                                 | 📈                                  | SQL                             |               |    |
|                                                                      | hierarchy1_id<br>character varying | hierarchy2_id<br>character varying | total_sales<br>double precision |               |    |
| 1                                                                    | [null]                             | [null]                             | 9856.880999999998               |               |    |
| 2                                                                    | H03                                | H0317                              | 2                               |               |    |
| 3                                                                    | H00                                | H0001                              | 617                             |               |    |
| 4                                                                    | H03                                | H0316                              | 0                               |               |    |
| 5                                                                    | H03                                | H0314                              | 190                             |               |    |
| Total rows: 23 of 23    Query complete 00:00:00.230    Ln 13, Col 14 |                                    |                                    |                                 |               |    |

- The hierarchy summary showed performance from category level down to individual product level.
- This supports product-range review and category-level decision-making.

## Module 2 Lesson 2: Sales Trends Over Time and Across Regions



The screenshot displays a database query editor with two tabs: 'Query' and 'Query History'. The 'Query' tab is active, showing an SQL query that calculates total sales by city and month. The query is as follows:

```
4 SUM(s.sales) AS total_sales
5 FROM
6   sales s
7 JOIN
8   store_cities c
9 ON
10  s.store_id = c.store_id
11 GROUP BY
12   CUBE (
13     c.city_id,
14     TO_CHAR(s."date", 'YYYY-MM')
```

Below the query editor, there are three tabs: 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table of results. The table has four columns: 'city\_id' (character varying), 'sales\_month' (text), 'total\_sales' (double precision), and an unnamed column. The results show two rows of data for city C002.

|   | city_id<br>character varying | sales_month<br>text | total_sales<br>double precision |  |
|---|------------------------------|---------------------|---------------------------------|--|
| 1 | C002                         | 2017-02             | 100                             |  |
| 2 | C002                         | 2017-03             | 59                              |  |

At the bottom of the 'Data Output' tab, the status bar indicates: 'Total rows: 102 of 102', 'Query complete 00:00:00.232', and 'Ln 18, Col 17'.

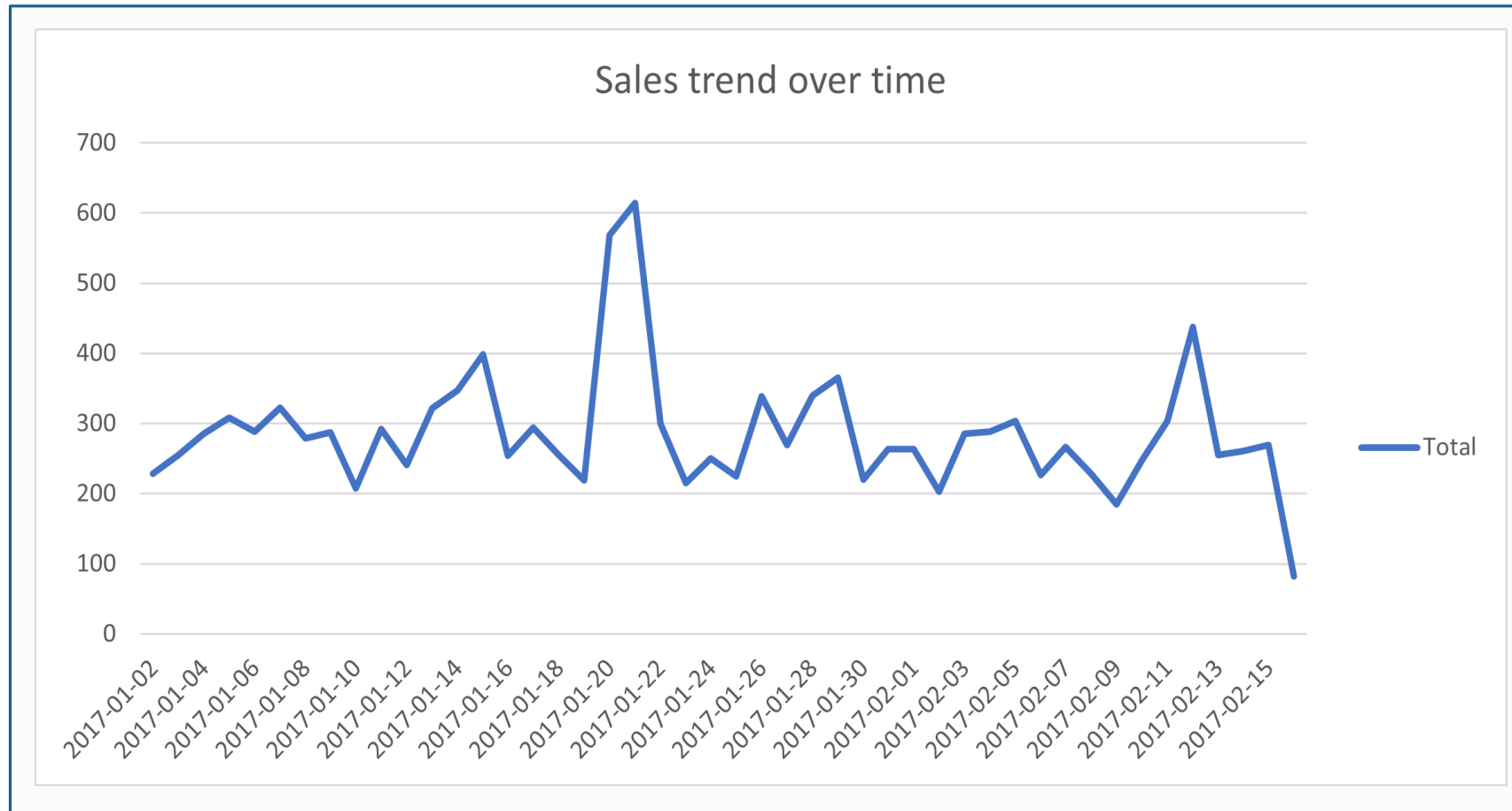
- The query identified monthly sales and revenue patterns by city or region.
- This supports demand planning and region-specific performance monitoring.

## Module 2 Results: Optimising Sales and Stock Management

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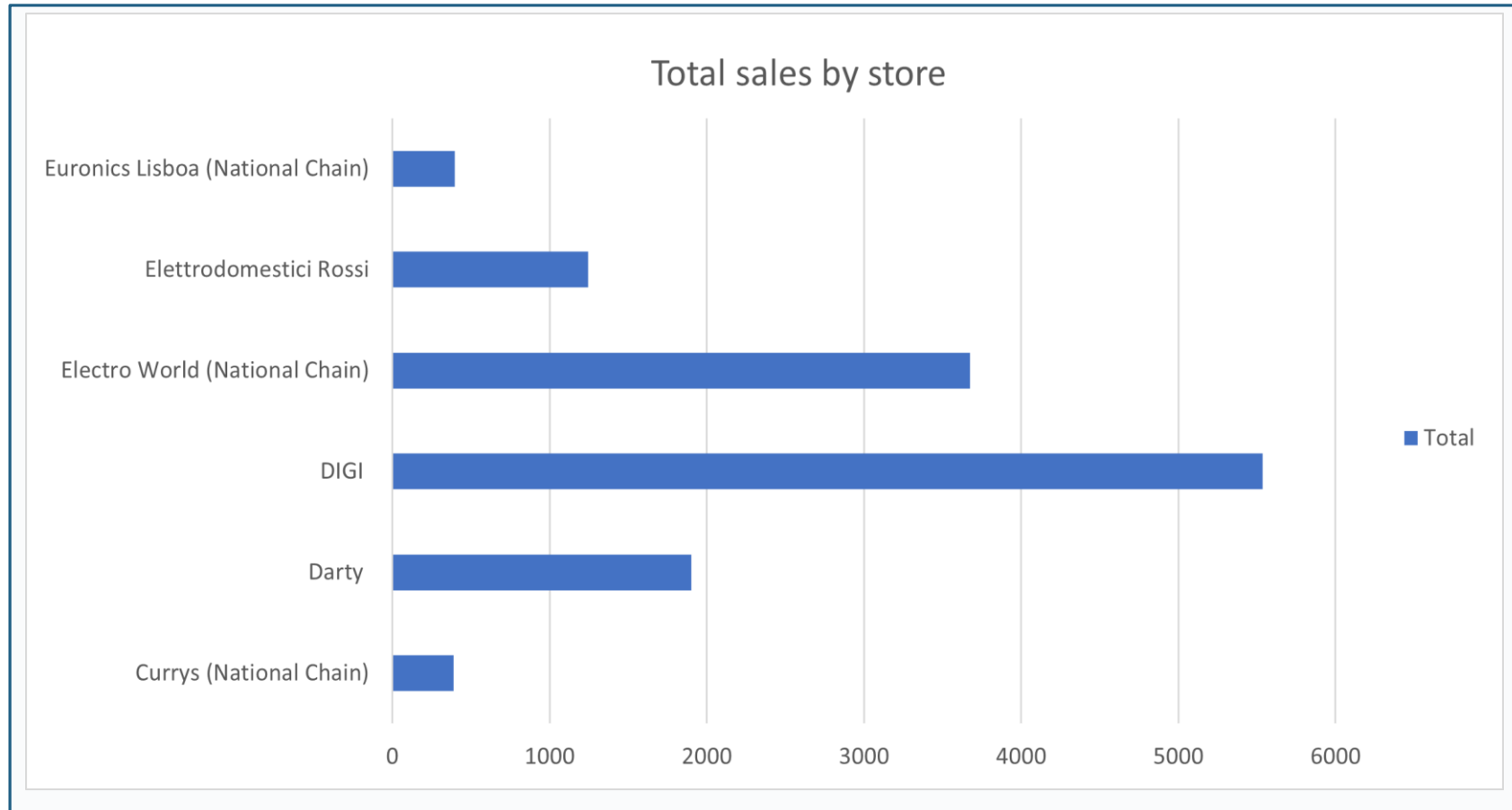
- ROLLUP outputs make it easier to compare detailed results with subtotals and overall totals.
- Hierarchy summaries show which product categories and subcategories are most important to sales and revenue.
- Time and region trend analysis helps identify where demand is increasing, decreasing or fluctuating.
- These insights support sales strategy by showing where to target promotions and where performance needs attention.
- They also support stock management by showing which stores, regions or products may need higher or lower inventory levels.

## Module 3 Lesson 1: Excel Line Chart



- Shows changes in sales or revenue across time.
- Helps identify upward trends, downward trends, fluctuations or seasonal behaviour.
- Useful for monitoring whether the business is improving or declining over the selected period.

## Module 3 Lesson 1: Excel Bar Chart



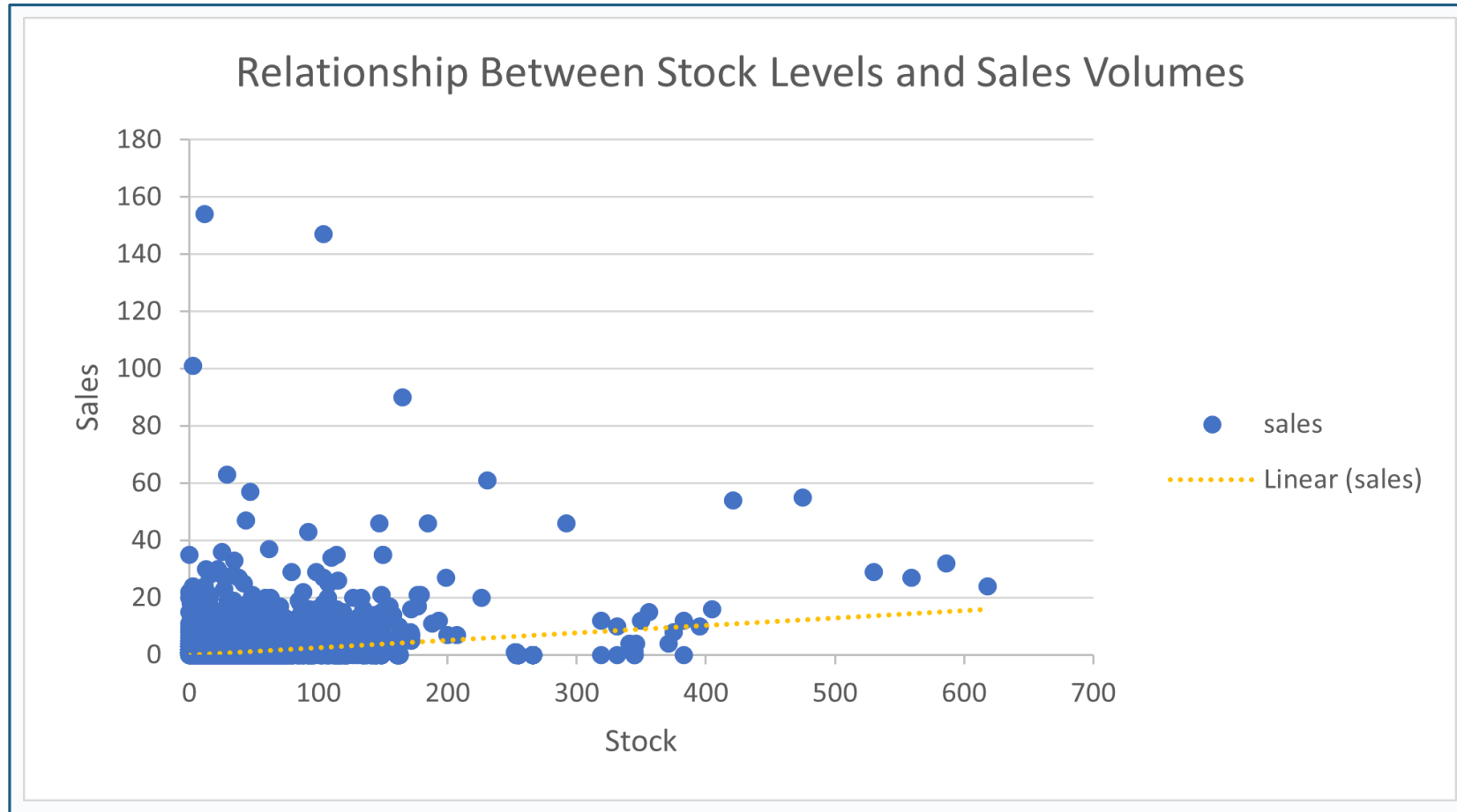
- Compares sales or revenue across categories.
- Makes top and bottom performers easier to identify.
- Supports decisions about where management attention, marketing or stock adjustment may be needed.

## Module 3 Lesson 1: Excel Sunburst Chart



- This chart visualises hierarchical product data such as category, subcategory and product.
- Shows how each product level contributes to the total.
- Useful for understanding product mix and identifying the categories that drive overall performance.

## Module 3 Lesson 1: Excel Scatter Plot



- This stock-sales analysis suggested a weak positive relationship: higher stock levels were sometimes associated with higher sales.
- The relationship was not strong, so other factors such as price, product type and customer demand are also likely to influence sales.

## Module 3 Lesson 1: Contribution of Excel Charts

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### Line chart

The line chart showed how sales or revenue changed over time, making it easier to identify trends, fluctuations and possible periods of stronger or weaker performance. This chart shows that **DIGI** generated the highest total sales, followed by **Electro World** and **Darty**. It helps identify the strongest and weakest performing stores

### Bar chart

Compares performance across categories such as cities, stores or products.

This chart shows that sales fluctuated over the period, with clear peaks in late January and mid-February. It helps identify periods of high and low sales activity.

### Sunburst

This chart shows how sales are distributed across different levels of the business, from cities to stores and products. It helps understand which areas and product groups contribute most to overall sales.

### Scatter plot

This chart shows a weak positive relationship between stock and sales. Most products had low-to-medium stock and sales, while a few outliers had much higher stock or sales values.

## Module 3 Lesson 2: Regression Analysis Output

A multiple regression analysis was conducted to estimate sales using date, stock and price as independent variables. The regression equation was: Predicted Sales =

A multiple regression analysis was conducted to estimate sales using date, stock and price as independent variables. The regression equation was: Predicted Sales = 14.10520159 - 0.000325058(sales\_date) + 0.027388943(stock) - 0.010628247(price). The overall model was statistically significant, as shown by the Significance F value. However, the R Square value was approximately 0.079, meaning the model explains only around 7.9% of the variation in sales. Stock had a positive and statistically significant relationship with sales, while price had a negative and statistically significant relationship with sales. Date was not statistically significant, suggesting that date alone does not meaningfully explain sales changes in this model.

|            | df    | SS          | MS          | F           | Significance F |
|------------|-------|-------------|-------------|-------------|----------------|
| Regression | 3     | 12382.51257 | 4127.50419  | 569.0372218 | 0              |
| Residual   | 19995 | 145033.4761 | 7.253487174 |             |                |
| Total      | 19998 | 157415.9886 |             |             |                |

|            | Coefficients | Standard Error | t Stat       | P-value     | Lower 95%    | Upper 95%    | Lower 95.0%  | Upper 95.0%  |
|------------|--------------|----------------|--------------|-------------|--------------|--------------|--------------|--------------|
| Intercept  | 14.10520159  | 63.25213094    | 0.222999627  | 0.823538075 | -109.8742019 | 138.0846051  | -109.8742019 | 138.0846051  |
| sales_date | -0.000325058 | 0.00147931     | -0.219736103 | 0.826078929 | -0.003224628 | 0.002574512  | -0.003224628 | 0.002574512  |
| stock      | 0.027388943  | 0.00070134     | 39.05228684  | 0           | 0.026014258  | 0.028763628  | 0.026014258  | 0.028763628  |
| price      | -0.010628247 | 0.001518868    | -6.997477383 | 2.68855E-12 | -0.013605354 | -0.007651139 | -0.013605354 | -0.007651139 |

- Dependent variable: Sales.
- Independent variables used in the model: sales date, stock and price.
- Regression equation from the analysis:  
$$\text{Predicted Sales} = 14.10520159 - 0.000325058(\text{sales\_date}) + 0.027388943(\text{stock}) - 0.010628247(\text{price})$$
- Stock had a positive coefficient; price had a negative coefficient; date had a very small negative coefficient.

## Module 3 Lesson 2: Interpretation of R-squared and P-values

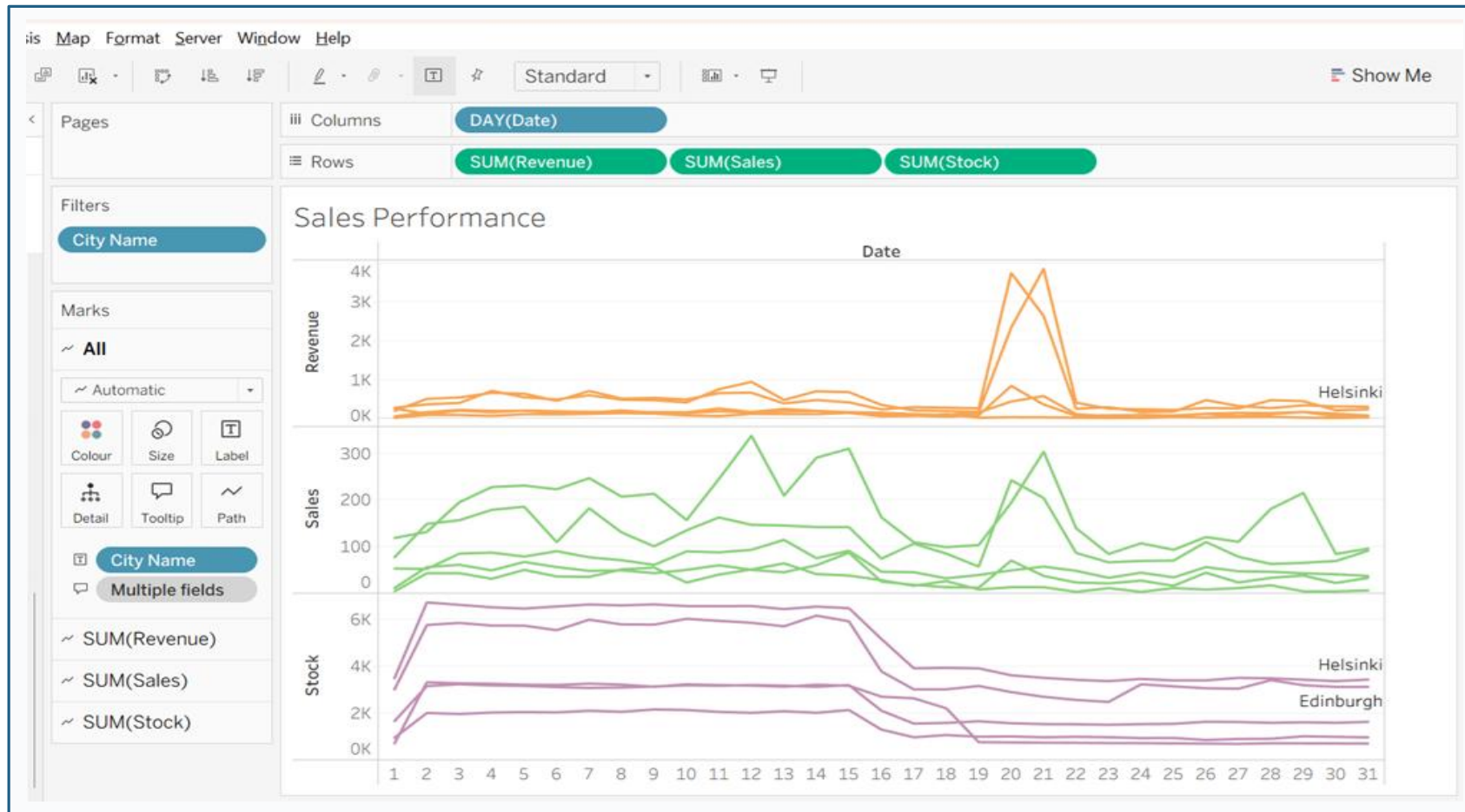
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The regression model has an R-squared value of approximately 0.079, meaning that sales date, stock and price explained around 7.9% of the variation in sales. This shows that the model has limited predictive power, although it still provides useful insight into the direction of relationships between the variables.

The overall model is statistically significant, as shown by the Significance F value. However, the individual p-values show that not all variables were meaningful predictors. Sales date has a high p-value of around 0.826, so it is not statistically significant. Stock is highly significant and had a positive coefficient, suggesting that higher stock levels were linked with higher sales. Price was also statistically significant and had a negative coefficient, suggesting that higher prices were linked with lower sales.

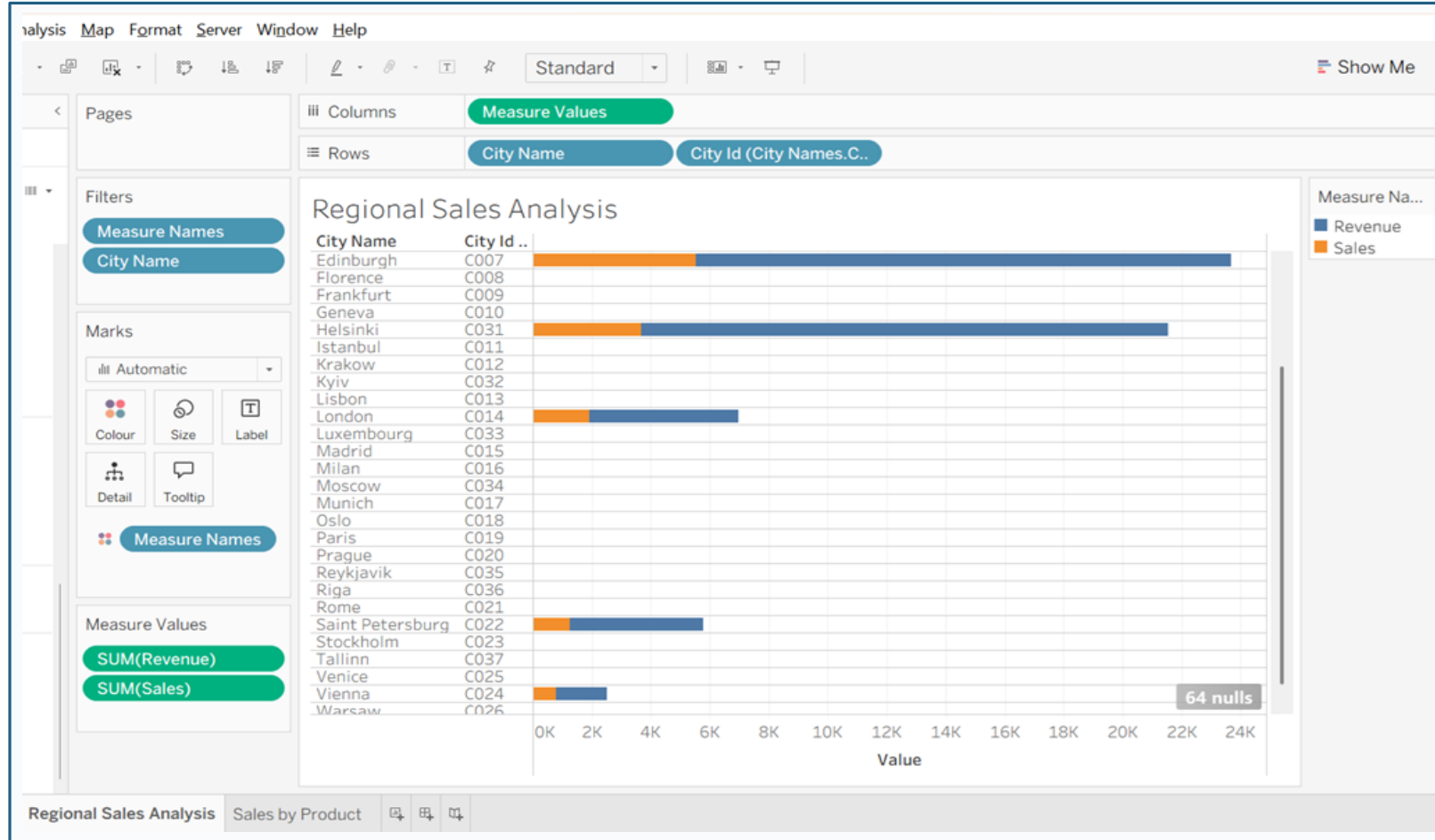
Overall, the regression suggests that stock and price were useful predictors of sales, while date was not. However, the low R-squared value indicates that other factors not included in the model also had a major influence on sales performance.

## Module 4 Lesson 1: Tableau Line Chart



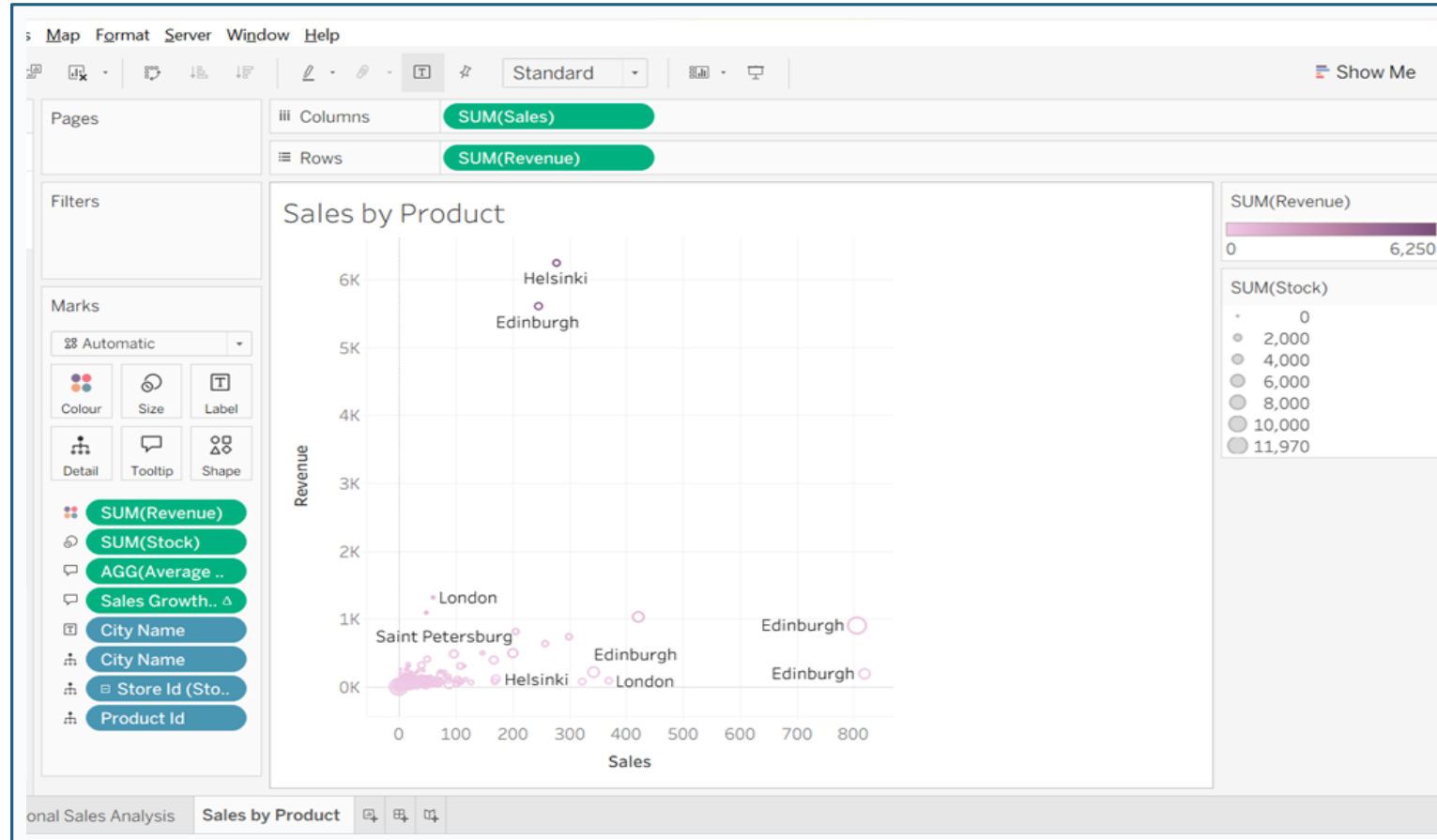
- Sales, revenue and stock trends over time.
- This chart makes increases, decreases and fluctuations easy to communicate.
- Supports quick understanding of time-based performance without reading detailed tables.

## Module 4 Lesson 1: Tableau Bar Chart



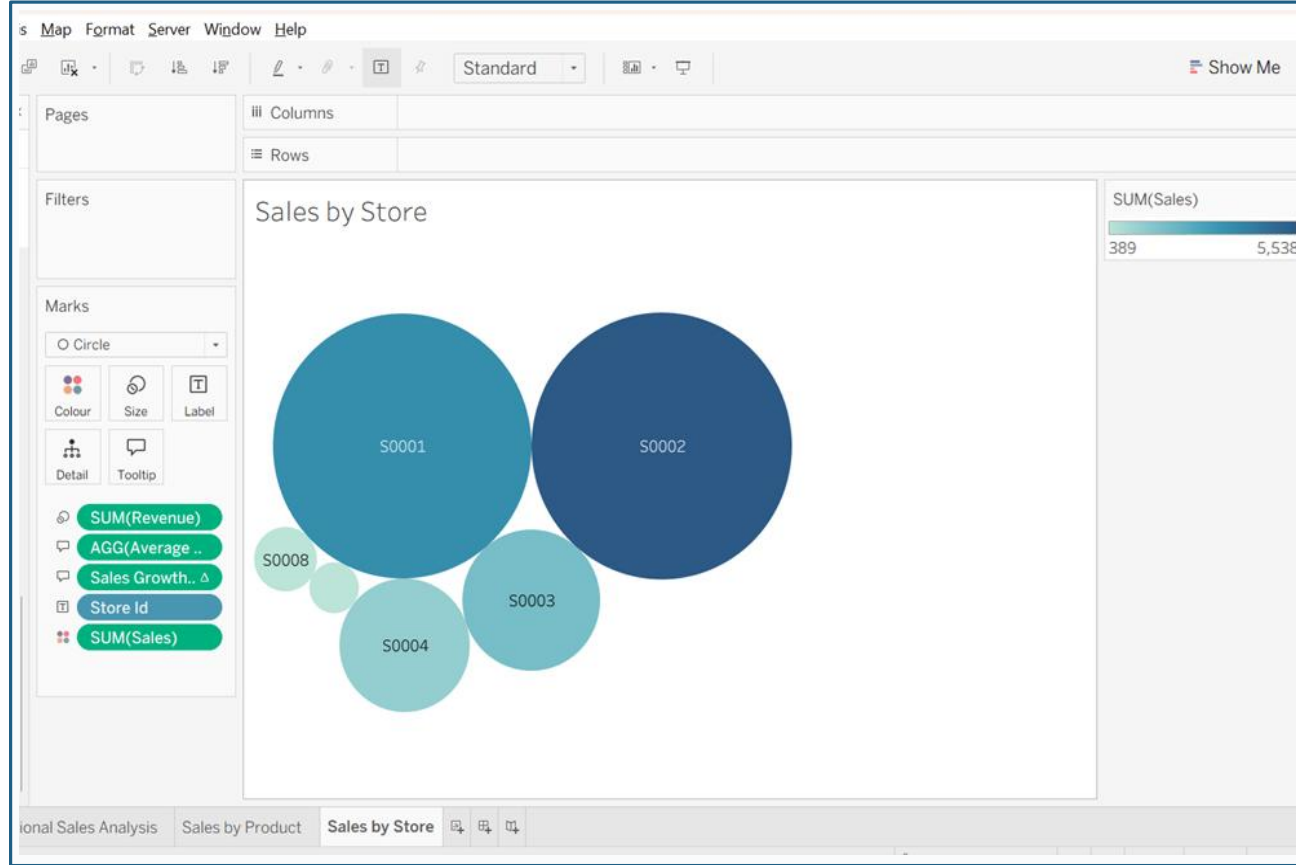
- Compares sales and revenue across cities.
- Helps identify top and bottom performers clearly.
- Useful for presenting ranking and comparison insights to business stakeholders.

## Module 4 Lesson 1: Tableau Scatter Plot



- Displays the relationship between sales and revenue at the product level.
- Helps identify clusters, outliers and possible associations.
- Supports further investigation into factors affecting sales performance.

## Module 4 Lesson 1: Tableau Packed Bubbles Chart



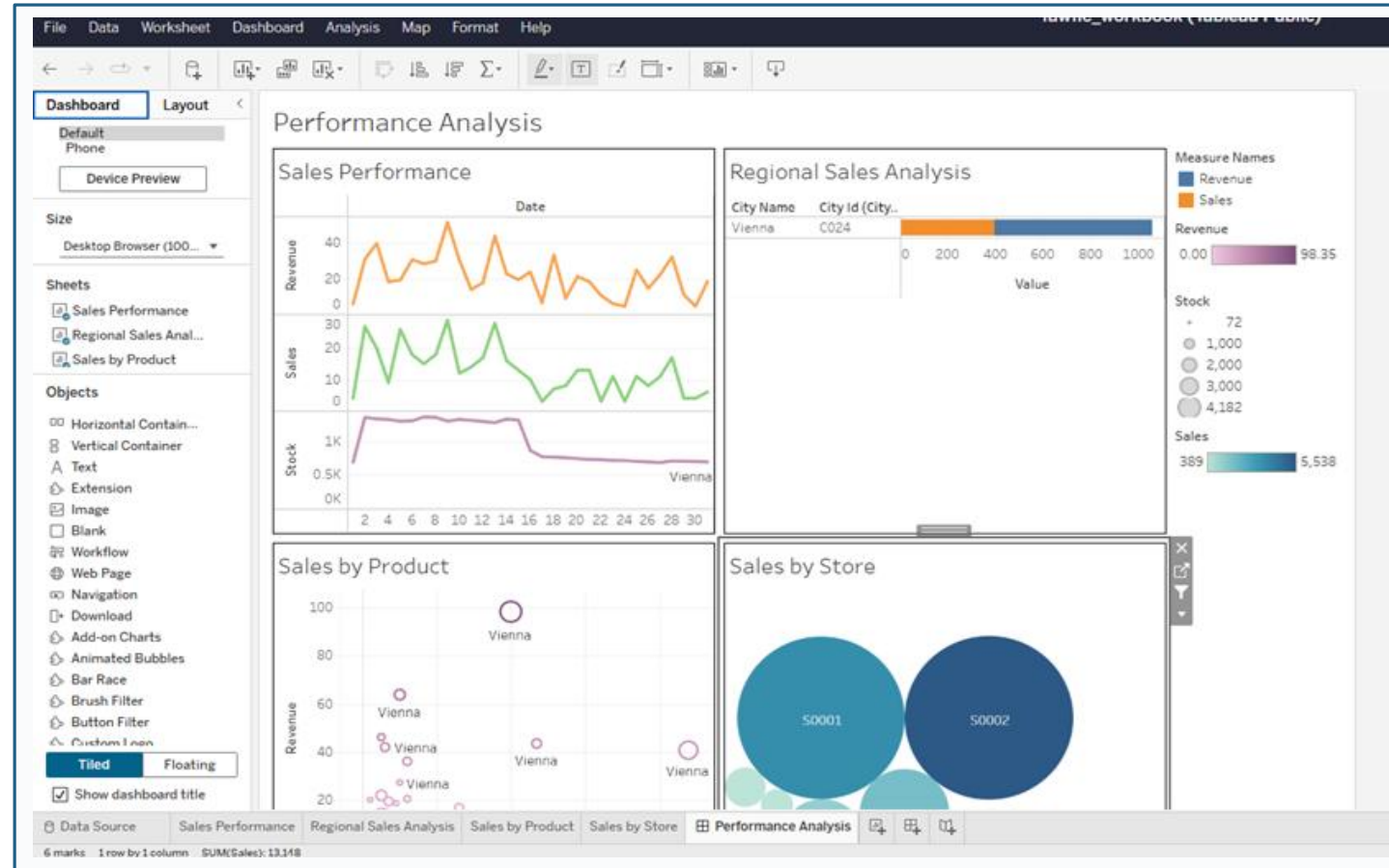
- The bubble size in this chart shows the relative size of sales at a store compared to others.
- Provides an engaging visual comparison between stores.
- Makes the largest contributors easy to identify at a glance.

## Module 4 Lesson 1: Communicating Insights with Tableau

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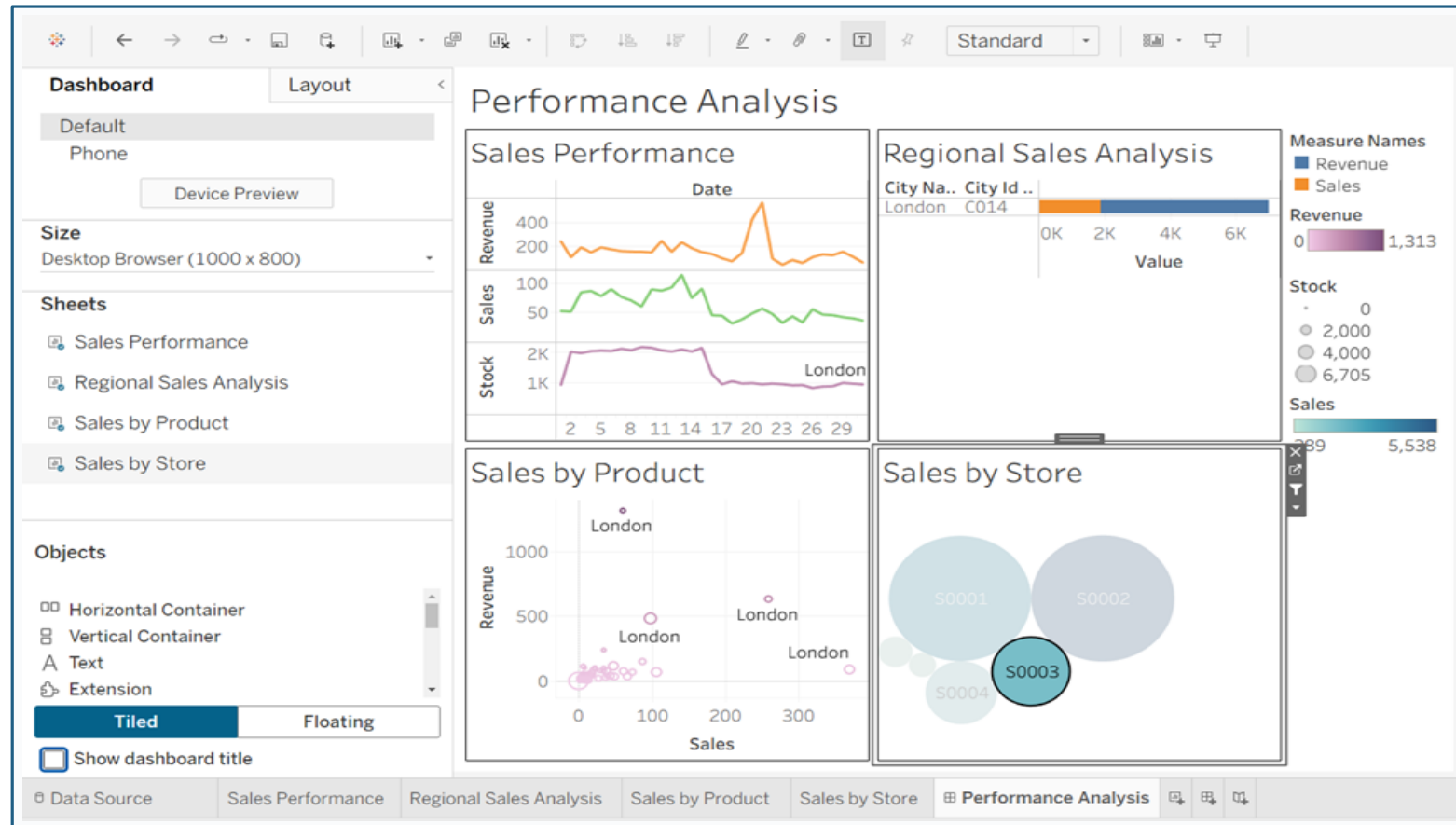
- Tableau visualisations made the results more accessible for business users.
- Line charts communicated trends, bar charts communicated comparisons, scatter plots communicated relationships and packed bubbles communicated relative contribution.
- The visuals reduced the need for stakeholders to interpret large tables manually.
- By presenting information graphically, Tableau helped turn technical analysis into clear business messages.

## Module 4 Lesson 2: Tableau Dashboard



- The dashboard combined multiple worksheets into one interactive analytical view.
- It brought together trend, comparison, relationship and contribution visuals.
- This allows users to monitor performance across several dimensions without moving between separate sheets.

## Module 4 Lesson 2: Filter Dashboard Action



- A filter action made the dashboard interactive.
- Selecting a store in “sales by store” chart filtered the other charts.
- This enabled focused investigation into specific business areas rather than only showing overall totals.

## Module 4 Lesson 2: Decision-Making Value of Dashboards

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- The dashboard supports decision-making by presenting sales performance, product contribution, regional differences and stock-sales relationships in one place.
- Interactive filtering allows managers to drill down into specific cities, stores or products.
- This helps users move from broad monitoring to targeted investigation.
- The dashboard can support decisions about stock allocation, product promotion, regional performance improvement and future sales monitoring.



Discussion

## Insights from the Analysis

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The analysis shows that sales performance is uneven across the retail chain. The Excel bar chart shows that DIGI achieved the highest total sales, followed by Electro World and Darty, while stores such as Currys and Euronics Lisboa had much lower sales. This suggests that some stores are contributing much more strongly to overall performance than others.

The sales trend chart shows that sales fluctuated over time, with noticeable peaks around late January and mid-February. This indicates that sales activity is not stable across the period and may be affected by timing, product availability, promotions or customer demand.

The product, city and store analysis also shows that sales are not evenly distributed across the business. The sunburst chart and Tableau visualisations show that some cities, stores and products contribute more heavily to total sales and revenue. This gives the retail chain a clearer view of where performance is strongest and where further investigation is needed.

The scatter plot and regression analysis suggest that stock has a positive relationship with sales, while price has a negative relationship with sales. However, the R-squared value of approximately 7.9% shows that stock, price and date explain only a small part of sales variation, meaning other factors also influence sales performance.

## Communication and Recommendations

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The presentation communicates the findings by using different analysis methods for different purposes. Excel pivot tables summarised sales, revenue and product performance. PostgreSQL queries provided structured analysis across tables, time periods and hierarchies. Excel charts made trends, comparisons and relationships easier to understand. Tableau dashboards brought the key findings together in an interactive format, allowing stakeholders to filter results by city, store or product.

Based on the insights, the retail chain should focus on strengthening high-performing stores such as DIGI while reviewing the reasons behind weaker store performance. Stock planning should be improved because the analysis suggests that higher stock levels are associated with higher sales. The business should also consider price sensitivity, as the regression results show a negative relationship between price and sales.

The retail chain should use dashboards regularly to monitor sales trends, identify underperforming locations, compare product performance and support stock allocation decisions. Further analysis should also include additional factors such as promotions, product categories, seasonality and store size to improve the accuracy of future sales forecasting.



# Conclusion

## Summary

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The analysis showed that the retail chain's sales performance varied across stores, cities, products and time periods. DIGI recorded the highest total sales, while other stores showed much lower performance. Sales also fluctuated over time, with noticeable peaks in late January and mid-February.

The Excel, PostgreSQL and Tableau analysis helped identify key patterns in sales, revenue, stock and product performance. The scatter plot and regression analysis suggested that higher stock levels were linked with higher sales, while higher prices were linked with lower sales. However, the low R-squared value showed that other factors also influenced sales performance.

Overall, the project demonstrated how business intelligence tools can turn raw retail data into useful insights. The findings can help the retail chain improve stock planning, monitor store and product performance, review weaker areas and make more informed business decisions.

### **In practical terms:**

- The project demonstrated the full business intelligence workflow: data cleaning > preparation > querying > analysis > visualisation > dashboard development.
- Excel supported data preparation, pivot table summaries, charts and regression analysis.
- PostgreSQL supported structured querying, aggregation, ROLLUP, hierarchy analysis and trend analysis.
- Tableau supported clear communication of insights through interactive visualisations and dashboards.
- The final analysis showed how retail data can support better sales strategy, stock management and operational decision-making.



# Appendix

# Appendix: Data cleaning – Task 1 (Before)

|    | A            | B          | C          | D       | E         | F       | G       | H              | I             | J              |
|----|--------------|------------|------------|---------|-----------|---------|---------|----------------|---------------|----------------|
| 1  | product_id ▾ | store_id ▾ | date ▾     | sales ▾ | revenue ▾ | stock ▾ | price ▾ | promo_type_1 ▾ | promo_bin_1 ▾ | promo_type_2 ▾ |
| 2  | P0001        | S0002      | 02/01/2017 | 0       | 0         | 8       | 6.25    | PR14           |               | PR03           |
| 3  | P0001        | S0012      | 02/01/2017 | 1       | 5.3       | 0       | 6.25    | PR14           |               | PR03           |
| 4  | P0001        | S0013      | 02/01/2017 | 2       | 10.59     | 0       | 6.25    | PR14           |               | PR03           |
| 5  | P0001        | S0023      | 02/01/2017 | 0       | 0         | 6       | 6.25    | PR14           |               | PR03           |
| 6  | P0001        | S0025      | 02/01/2017 | 0       | 0         | 1       | 6.25    | PR14           |               | PR03           |
| 7  | P0001        | S0027      | 02/01/2017 | 0       | 0         | 7       | 6.25    | PR14           |               | PR03           |
| 8  | P0001        | S0040      | 02/01/2017 | 0       | 0         | 19      | 6.25    | PR14           |               | PR03           |
| 9  | P0001        | S0049      | 02/01/2017 | 0       | 0         | 8       | 6.25    | PR14           |               | PR03           |
| 10 | P0001        | S0050      | 02/01/2017 | 0       | 0         | 5       | 6.25    | PR14           |               | PR03           |
| 11 | P0001        | S0051      | 02/01/2017 | 0       | 0         | 6       | 6.25    | PR14           |               | PR03           |
| 12 | P0001        | S0055      | 02/01/2017 | 0       | 0         | 6       | 6.25    | PR14           |               | PR03           |
| 13 | P0001        | S0056      | 02/01/2017 | 1       | 5.3       | 6       | 6.25    | PR14           |               | PR03           |
| 14 | P0001        | S0062      | 02/01/2017 | 0       | 0         | 2       | 6.25    | PR14           |               | PR03           |
| 15 | P0001        | S0063      | 02/01/2017 | 0       | 0         | 7       | 6.25    | PR14           |               | PR03           |
| 16 | P0001        | S0066      | 02/01/2017 | 0       | 0         | 1       | 6.25    | PR14           |               | PR03           |
| 17 | P0001        | S0078      | 02/01/2017 | 0       | 0         | 6       | 6.25    | PR14           |               | PR03           |
| 18 | P0001        | S0082      | 02/01/2017 | 0       | 0         | 4       | 6.25    | PR14           |               | PR03           |
| 19 | P0001        | S0083      | 02/01/2017 | 0       | 0         | 10      | 6.25    | PR14           |               | PR03           |
| 20 | P0001        | S0087      | 02/01/2017 | 0       | 0         | 10      | 6.25    | PR14           |               | PR03           |

# Appendix: Data cleaning – Task 1 (After)

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|    | A          | B        | C          | D     | E       | F     | G     | H            | I           | J        |
|----|------------|----------|------------|-------|---------|-------|-------|--------------|-------------|----------|
| 1  | product_id | store_id | date       | sales | revenue | stock | price | promo_type_1 | promo_bin_1 | promo_ty |
| 2  | P0001      | S0002    | 02/01/2017 | £0.00 | 0.00    | 8.00  | £6.25 | PR14         |             | PR03     |
| 3  | P0001      | S0012    | 02/01/2017 | £1.00 | 5.30    | 0.00  | £6.25 | PR14         |             | PR03     |
| 4  | P0001      | S0013    | 02/01/2017 | £2.00 | 10.59   | 0.00  | £6.25 | PR14         |             | PR03     |
| 5  | P0001      | S0023    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03     |
| 6  | P0001      | S0025    | 02/01/2017 | £0.00 | 0.00    | 1.00  | £6.25 | PR14         |             | PR03     |
| 7  | P0001      | S0027    | 02/01/2017 | £0.00 | 0.00    | 7.00  | £6.25 | PR14         |             | PR03     |
| 8  | P0001      | S0040    | 02/01/2017 | £0.00 | 0.00    | 19.00 | £6.25 | PR14         |             | PR03     |
| 9  | P0001      | S0049    | 02/01/2017 | £0.00 | 0.00    | 8.00  | £6.25 | PR14         |             | PR03     |
| 10 | P0001      | S0051    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03     |
| 11 | P0001      | S0055    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03     |
| 12 | P0001      | S0056    | 02/01/2017 | £1.00 | 5.30    | 6.00  | £6.25 | PR14         |             | PR03     |
| 13 | P0001      | S0062    | 02/01/2017 | £0.00 | 0.00    | 2.00  | £6.25 | PR14         |             | PR03     |
| 14 | P0001      | S0063    | 02/01/2017 | £0.00 | 0.00    | 7.00  | £6.25 | PR14         |             | PR03     |
| 15 | P0001      | S0066    | 02/01/2017 | £0.00 | 0.00    | 1.00  | £6.25 | PR14         |             | PR03     |
| 16 | P0001      | S0078    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03     |
| 17 | P0001      | S0082    | 02/01/2017 | £0.00 | 0.00    | 4.00  | £6.25 | PR14         |             | PR03     |
| 18 | P0001      | S0083    | 02/01/2017 | £0.00 | 0.00    | 10.00 | £6.25 | PR14         |             | PR03     |
| 19 | P0001      | S0087    | 02/01/2017 | £0.00 | 0.00    | 10.00 | £6.25 | PR14         |             | PR03     |
| 20 | P0001      | S0094    | 02/01/2017 | £0.00 | 0.00    | 6.00  | £6.25 | PR14         |             | PR03     |

# Appendix: Data cleaning – Task 2 (Before)

|                                                                                       | A            | B                | C               | D               | E            | F               | G               | H               | I               | J               | K |
|---------------------------------------------------------------------------------------|--------------|------------------|-----------------|-----------------|--------------|-----------------|-----------------|-----------------|-----------------|-----------------|---|
| 1                                                                                     | product_id ▾ | product_length ▾ | product_depth ▾ | product_width ▾ | cluster_id ▾ | hierarchy1_id ▾ | hierarchy2_id ▾ | hierarchy3_id ▾ | hierarchy4_id ▾ | hierarchy5_id ▾ |   |
| 2                                                                                     | P0000        | 5                | 20              | 12              |              | H00             | H0004           | H000401         | H00040105       | H0004010534     |   |
| 3                                                                                     | P0001        | 13.5             | 22              | 20              | cluster_5    | H01             | H0105           | H010501         | H01050100       | H0105010006     |   |
| 4                                                                                     | P0002        | 22               | 40              | 22              | cluster_0    | H03             | H0315           | H031508         | H03150800       | H0315080028     |   |
| 5                                                                                     | P0004        | 2                | 13              | 4               | cluster_3    | H03             | H0314           | H031405         | H03140500       | H0314050003     |   |
| 6                                                                                     | P0005        | 16               | 30              | 16              | cluster_9    | H03             | H0312           | H031211         | H03121109       | H0312110917     |   |
| 7                                                                                     | P0006        | 8.5              | 15              | 15              | cluster_0    | H03             | H0316           | H031608         | H03160817       | H0316081708     |   |
| 8                                                                                     | P0007        | 2                | 22              | 9.5             | cluster_4    | H03             | H0313           | H031305         | H03130519       | H0313051904     |   |
| 9                                                                                     | P0008        | 5                | 16              | 5               | cluster_0    | H00             | H0000           | H000004         | H00000400       | H0000040017     |   |
| 0                                                                                     | P0009        | 5                | 18              | 14              | cluster_6    | H00             | H0002           | H000201         | H00020100       | H0002010012     |   |
| 1                                                                                     | P0010        | 2                | 22              | 3               | cluster_0    | H01             | H0108           | H010801         | H01080109       | H0108010917     |   |
| 2                                                                                     | P0011        | 9                | 22              | 15              | cluster_0    | H03             | H0315           | H031506         | H03150612       | H0315061202     |   |
| 3                                                                                     | P0012        | 10               | 20              | 12.3            | cluster_0    | H01             | H0107           | H010702         | H01070210       | H0107021001     |   |
| 4                                                                                     | P0013        | 7.1              | 27.5            | 13.7            |              | H01             | H0105           | H010503         | H01050309       | H0105030906     |   |
| 5                                                                                     | P0014        | 32.5             | 6               | 33              | cluster_0    | H03             | H0313           | H031302         | H03130205       | H0313020501     |   |
| 6                                                                                     | P0015        | 10               | 33              | 10              | cluster_1    | H00             | H0000           | H000003         | H00000309       | H0000030901     |   |
| 7                                                                                     | P0016        | 2                | 18              | 8               | cluster_0    | H00             | H0004           | H000401         | H00040105       | H0004010512     |   |
| 8                                                                                     | P0017        | 8                | 28              | 8               | cluster_6    | H00             | H0000           | H000003         | H00000310       | H0000031001     |   |
| 9                                                                                     | P0018        | 1                | 14              | 11              | cluster_4    | H00             | H0003           | H000316         | H00031609       | H0003160922     |   |
| 0                                                                                     | P0019        | 1.5              | 42              | 32              | cluster_0    | H03             | H0312           | H031201         | H03120100       | H0312010004     |   |
| ◀ ▶ sales product_hierarchy store_cities store_names city_names product_names (+) ⋮ ◀ |              |                  |                 |                 |              |                 |                 |                 |                 |                 |   |

# Appendix: Data cleaning – Task 2 (After)

|                                                                                                                                                                                                     | A            | B                | C               | D               | E            | F               | G               | H               | I               | J               | K |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|-----------------|-----------------|--------------|-----------------|-----------------|-----------------|-----------------|-----------------|---|
| 1                                                                                                                                                                                                   | product_id ▾ | product_length ▾ | product_depth ▾ | product_width ▾ | cluster_id ▾ | hierarchy1_id ▾ | hierarchy2_id ▾ | hierarchy3_id ▾ | hierarchy4_id ▾ | hierarchy5_id ▾ |   |
| 2                                                                                                                                                                                                   | P0001        | 13.50            | 22.00           | 20.00           | cluster_5    | H01             | H0105           | H010501         | H01050100       | H0105010006     |   |
| 3                                                                                                                                                                                                   | P0002        | 22.00            | 40.00           | 22.00           | cluster_0    | H03             | H0315           | H031508         | H03150800       | H0315080028     |   |
| 4                                                                                                                                                                                                   | P0004        | 2.00             | 13.00           | 4.00            | cluster_3    | H03             | H0314           | H031405         | H03140500       | H0314050003     |   |
| 5                                                                                                                                                                                                   | P0005        | 16.00            | 30.00           | 16.00           | cluster_9    | H03             | H0312           | H031211         | H03121109       | H0312110917     |   |
| 6                                                                                                                                                                                                   | P0006        | 8.50             | 15.00           | 15.00           | cluster_0    | H03             | H0316           | H031608         | H03160817       | H0316081708     |   |
| 7                                                                                                                                                                                                   | P0007        | 2.00             | 22.00           | 9.50            | cluster_4    | H03             | H0313           | H031305         | H03130519       | H0313051904     |   |
| 8                                                                                                                                                                                                   | P0008        | 5.00             | 16.00           | 5.00            | cluster_0    | H00             | H0000           | H000004         | H00000400       | H0000040017     |   |
| 9                                                                                                                                                                                                   | P0009        | 5.00             | 18.00           | 14.00           | cluster_6    | H00             | H0002           | H000201         | H00020100       | H0002010012     |   |
| 10                                                                                                                                                                                                  | P0010        | 2.00             | 22.00           | 3.00            | cluster_0    | H01             | H0108           | H010801         | H01080109       | H0108010917     |   |
| 11                                                                                                                                                                                                  | P0011        | 9.00             | 22.00           | 15.00           | cluster_0    | H03             | H0315           | H031506         | H03150612       | H0315061202     |   |
| 12                                                                                                                                                                                                  | P0012        | 10.00            | 20.00           | 12.30           | cluster_0    | H01             | H0107           | H010702         | H01070210       | H0107021001     |   |
| 13                                                                                                                                                                                                  | P0014        | 32.50            | 6.00            | 33.00           | cluster_0    | H03             | H0313           | H031302         | H03130205       | H0313020501     |   |
| 14                                                                                                                                                                                                  | P0015        | 10.00            | 33.00           | 10.00           | cluster_1    | H00             | H0000           | H000003         | H00000309       | H0000030901     |   |
| 15                                                                                                                                                                                                  | P0016        | 2.00             | 18.00           | 8.00            | cluster_0    | H00             | H0004           | H000401         | H00040105       | H0004010512     |   |
| 16                                                                                                                                                                                                  | P0017        | 8.00             | 28.00           | 8.00            | cluster_6    | H00             | H0000           | H000003         | H00000310       | H0000031001     |   |
| 17                                                                                                                                                                                                  | P0018        | 1.00             | 14.00           | 11.00           | cluster_4    | H00             | H0003           | H000316         | H00031609       | H0003160922     |   |
| 18                                                                                                                                                                                                  | P0019        | 1.50             | 42.00           | 32.00           | cluster_0    | H03             | H0312           | H031201         | H03120100       | H0312010004     |   |
| 19                                                                                                                                                                                                  | P0020        | 5.00             | 28.00           | 16.00           | cluster_0    | H03             | H0311           | H031109         | H03110901       | H0311090101     |   |
| 20                                                                                                                                                                                                  | P0021        | 8.00             | 26.00           | 23.00           | cluster_0    | H03             | H0313           | H031302         | H03130210       | H0313021002     |   |
| <div><div>◀ ▶</div><div>sales</div><div>product_hierarchy</div><div>store_cities</div><div>store_names</div><div>city_names</div><div>product_names</div><div>⊕</div><div>⋮</div><div>◀</div></div> |              |                  |                 |                 |              |                 |                 |                 |                 |                 |   |

# Appendix: Cleaned data – Task 3

|                                                                            | A          | B              | C            | D         | E | F | G | H |
|----------------------------------------------------------------------------|------------|----------------|--------------|-----------|---|---|---|---|
| 1                                                                          | store_id ▾ | storetype_id ▾ | store_size ▾ | city_id ▾ |   |   |   |   |
| 2                                                                          | S0091      | ST04           | 19.00        | C013      |   |   |   |   |
| 3                                                                          | S0012      | ST04           | 28.00        | C005      |   |   |   |   |
| 4                                                                          | S0045      | ST04           | 17.00        | C008      |   |   |   |   |
| 5                                                                          | S0032      | ST03           | 14.00        | C019      |   |   |   |   |
| 6                                                                          | S0027      | ST04           | 24.00        | C022      |   |   |   |   |
| 7                                                                          | S0088      | ST04           | 20.00        | C009      |   |   |   |   |
| 8                                                                          | S0095      | ST02           | 44.00        | C014      |   |   |   |   |
| 9                                                                          | S0055      | ST04           | 24.00        | C014      |   |   |   |   |
| 10                                                                         | S0099      | ST03           | 14.00        | C014      |   |   |   |   |
| 11                                                                         | S0078      | ST04           | 19.00        | C036      |   |   |   |   |
| 12                                                                         | S0006      | ST03           | 8.00         | C024      |   |   |   |   |
| 13                                                                         | S0135      | ST03           | 16.00        | C035      |   |   |   |   |
| 14                                                                         | S0066      | ST04           | 47.00        | C033      |   |   |   |   |
| 15                                                                         | S0106      | ST04           | 21.00        | C031      |   |   |   |   |
| 16                                                                         | S0112      | ST04           | 46.00        | C031      |   |   |   |   |
| 17                                                                         | S0059      | ST03           | 15.00        | C014      |   |   |   |   |
| 18                                                                         | S0129      | ST04           | 18.00        | C021      |   |   |   |   |
| 19                                                                         | S0116      | ST03           | 15.00        | C031      |   |   |   |   |
| 20                                                                         | S0098      | ST03           | 15.00        | C022      |   |   |   |   |
| ◀ ▶ sales product_hierarchy store_cities store_names city_names product_na |            |                |              |           |   |   |   |   |
| Ready Accessibility: Investigate                                           |            |                |              |           |   |   |   |   |

## Appendix: Database setup query

```
> theia@theiadocker-ahmadnorband: /home/project X

theia@theiadocker-ahmadnorband:/home/project$ export PGPASSWORD=pCSWDNXa1IPNVH0SyuTcA
Gr8; psql --host 172.21.171.219 -p 5432 -U postgres
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1), server 13.2)
Type "help" for help.

postgres=# CREATE DATABASE BICapstone
postgres=# \connect bicapstone
connection to server at "172.21.171.219", port 5432 failed: FATAL:  database "bicapstone" does not exist
Previous connection kept
postgres=# \connect BICapstone
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1), server 13.2)
You are now connected to database "BICapstone" as user "postgres".
BICapstone=# \include BI-dump.sql
SET
SET
SET
```

# Appendix: Data upload test on city\_names

Query

Query History

Scratch Pad X

1

SELECT \* FROM city\_names LIMIT 5;

Data Output

Messages

Notifications

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SQL

|   | city_id<br>[PK] character varying (255) | city_name<br>character varying (255) |
|---|-----------------------------------------|--------------------------------------|
| 1 | C001                                    | Amsterdam                            |
| 2 | C002                                    | Berlin                               |
| 3 | C003                                    | Barcelona                            |
| 4 | C004                                    | Budapest                             |
| 5 | C005                                    | Copenhagen                           |

Total rows: 5 of 5

Query complete 00:00:00.249

Ln 1, Col 25

# Appendix: Data upload test query on product\_names

Query

Query History

Scratch Pad X

1 SELECT \* FROM product\_names LIMIT 5;

Data Output

Messages

Notifications

SQL

|   | product_id<br>[PK] character varying | product_name<br>character varying (255) |
|---|--------------------------------------|-----------------------------------------|
| 1 | P0000                                | Bottom Freezer Refrigerator             |
| 2 | P0001                                | Side-by-Side Refrigerator               |
| 3 | P0002                                | Mini Fridge                             |
| 4 | P0004                                | Beverage Center                         |
| 5 | P0005                                | Wine Cooler                             |

Total rows: 5 of 5    Query complete 00:00:00.243    Ln 1, Col 28

# Appendix: Data upload test query on store\_cities

Query

Query History

Scratch Pad

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1

SELECT \* FROM store\_cities LIMIT 5;

Data Output

Messages

Notifications

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SQL

|   | store_id<br>character varying | storetype_id<br>character varying | store_size<br>bigint | city_id<br>character varying |
|---|-------------------------------|-----------------------------------|----------------------|------------------------------|
| 1 | S0091                         | ST04                              | 19                   | C013                         |
| 2 | S0012                         | ST04                              | 28                   | C005                         |
| 3 | S0045                         | ST04                              | 17                   | C008                         |
| 4 | S0032                         | ST03                              | 14                   | C019                         |
| 5 | S0027                         | ST04                              | 24                   | C022                         |

Total rows: 5 of 5    Query complete 00:00:00.290    Ln 1, Col 27

# Appendix: Data upload test query on store\_names

Open File  
Ctrl O

Query History

Scratch Pad X

SELECT \* FROM store\_names LIMIT 5;

Data Output

Messages

Notifications

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SQL

|   | store_id<br>[PK] character varying | store_name<br>character varying |
|---|------------------------------------|---------------------------------|
| 1 | S0005                              | MediaMarkt(National Chain)      |
| 2 | S0036                              | Saturn                          |
| 3 | S0104                              | Euronics                        |
| 4 | S0068                              | FNAC                            |
| 5 | S0086                              | Darty                           |

Total rows: 5 of 5    Query complete 00:00:00.258    Ln 1, Col 26

# Appendix: Data upload test query on sales

Query

Query History

Scratch Pad

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1

SELECT \* FROM sales LIMIT 5;

Data Output

Messages

Notifications

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SQL

|   | product_id<br>character varying | store_id<br>character varying | date<br>date | sales<br>double precision | revenue<br>double precision | stock<br>double precision | price<br>double prec |
|---|---------------------------------|-------------------------------|--------------|---------------------------|-----------------------------|---------------------------|----------------------|
| 1 | P0001                           | S0002                         | 2017-02-01   | 0                         | 0                           | 8                         |                      |
| 2 | P0001                           | S0012                         | 2017-02-01   | 1                         | 5.3                         | 0                         |                      |
| 3 | P0001                           | S0013                         | 2017-02-01   | 2                         | 10.59                       | 0                         |                      |
| 4 | P0001                           | S0023                         | 2017-02-01   | 0                         | 0                           | 6                         |                      |
| 5 | P0001                           | S0025                         | 2017-02-01   | 0                         | 0                           | 1                         |                      |

Total rows: 5 of 5

Query complete 00:00:00.265

Ln 1, Col 21

# Appendix: Joining datasets using SQL

Query

Query History

1

2

3

4

5

6

7

SELECT

hierarchy1\_id, hierarchy2\_id,

SUM(s.sales) AS total\_sales

FROM

sales s

JOIN

product\_hierarchy p

Data Output

Messages

Notifications

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SQL

|   | hierarchy1_id<br>character varying | hierarchy2_id<br>character varying | total_sales<br>double precision |
|---|------------------------------------|------------------------------------|---------------------------------|
| 1 | [null]                             | [null]                             | 9856.880999999998               |
| 2 | H03                                | H0317                              | 2                               |
| 3 | H00                                | H0001                              | 617                             |
| 4 | H03                                | H0316                              | 0                               |
| 5 | H03                                | H0314                              | 190                             |

Total rows: 23 of 23    Query complete 00:00:00.230    Ln 13, Col 14

# Appendix: Calculating subtotals using Cube in SQL

Query

Query History

4

SUM(s.sales) AS total\_sales

5

FROM

6

sales s

7

JOIN

8

store\_cities c

9

ON

10

s.store\_id = c.store\_id

11

GROUP BY

12

CUBE (

13

c.city\_id,

14

TO\_CHAR(s."date", 'YYYY-MM')

Data Output

Messages

Notifications

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▼

SQL

|   | city_id<br>character varying | sales_month<br>text | total_sales<br>double precision |
|---|------------------------------|---------------------|---------------------------------|
| 1 | C002                         | 2017-02             | 100                             |
| 2 | C002                         | 2017-03             | 59                              |

Total rows: 102 of 102

Query complete 00:00:00.232

Ln 18, Col 17

# Appendix: Data consolidated from multiple files for analysis

FileHomeInsertDrawPage LayoutFormulasDataReviewViewDeveloperHelpPower PivotTable DesignQuery

Get DataGet Data >RefreshAllQueries & ConnectionsPropertiesEdit Links

StocksCurrencies

SortFilterClearReapplyAdvanced

Text to ColumnsData Tools

What-If AnalysisForecast SheetOutlineData AnalysisSolver

Share

Get & Transform DataQueries & ConnectionsData TypesSort & FilterData ToolsForecastAnalyze

A1fxproduct\_id

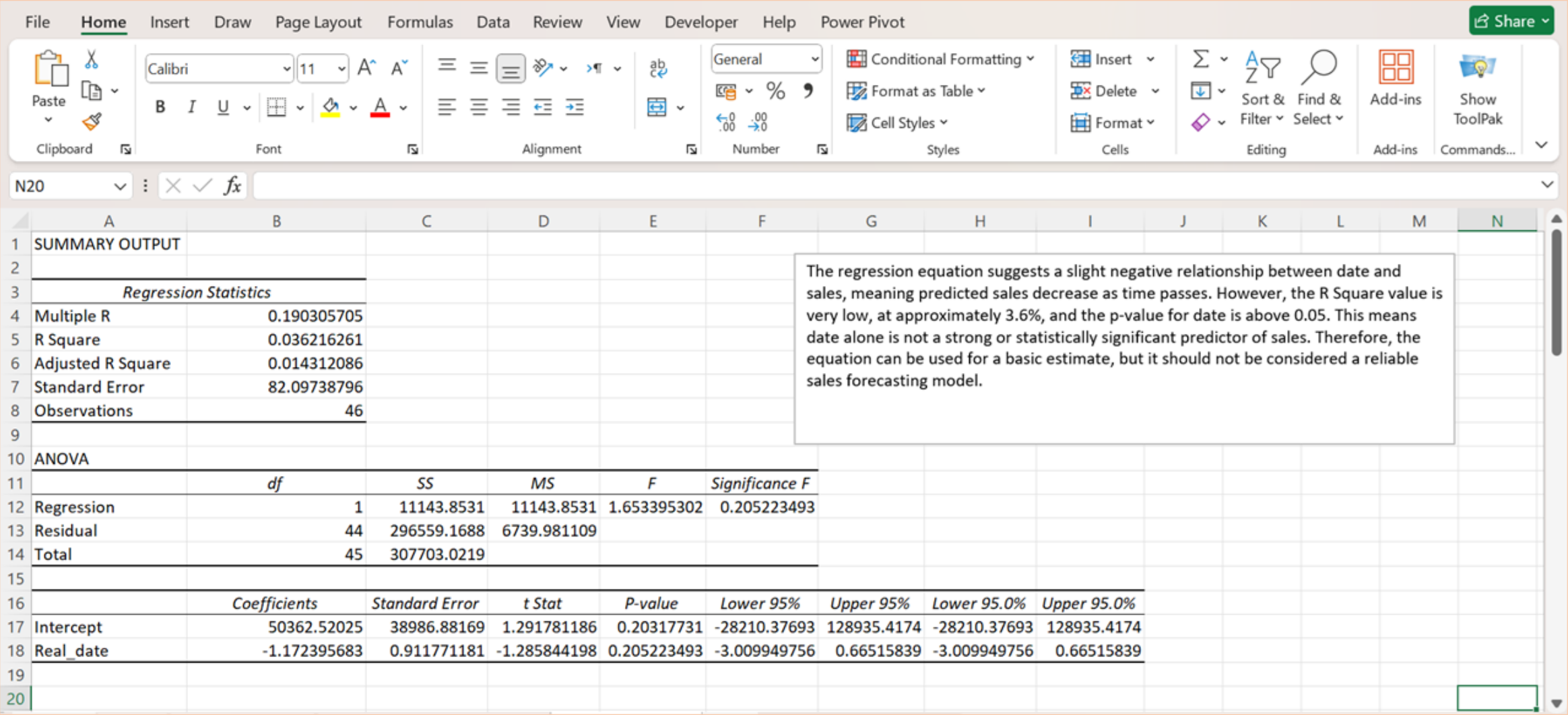
|    | A          | B                 | C        | D                              | E       | F         | G     | H       | I     | J     | K    | L          | M           |
|----|------------|-------------------|----------|--------------------------------|---------|-----------|-------|---------|-------|-------|------|------------|-------------|
| 1  | product_id | product_name      | store_id | store_name                     | city_id | city_name | sales | revenue | stock | price | date | promo_type | promo_bin_1 |
| 2  | P0005      | Wine Cooler       | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 7     | 33.9  |      | PR14       |             |
| 3  | P0005      | Wine Cooler       | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 7     | 33.9  |      | PR14       |             |
| 4  | P0005      | Wine Cooler       | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 7     | 33.9  |      | PR14       |             |
| 5  | P0005      | Wine Cooler       | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 7     | 33.9  |      | PR14       |             |
| 6  | P0011      | Induction Cooktop | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 10    | 49.9  |      | PR14       |             |
| 7  | P0011      | Induction Cooktop | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 10    | 49.9  |      | PR14       |             |
| 8  | P0011      | Induction Cooktop | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 10    | 49.9  |      | PR14       |             |
| 9  | P0011      | Induction Cooktop | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 10    | 49.9  |      | PR14       |             |
| 10 | P0015      | Convection Oven   | S0001    | Electro World (National Chain) | C031    | Helsinki  | 1     | 2.41    | 20    | 2.6   |      | PR14       |             |
| 11 | P0015      | Convection Oven   | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 20    | 2.6   |      | PR14       |             |
| 12 | P0015      | Convection Oven   | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 21    | 2.6   |      | PR14       |             |
| 13 | P0015      | Convection Oven   | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 21    | 2.6   |      | PR10       | low         |
| 14 | P0017      | Pizza Oven        | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 13    | 1.49  |      | PR14       |             |
| 15 | P0017      | Pizza Oven        | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 13    | 1.49  |      | PR14       |             |
| 16 | P0017      | Pizza Oven        | S0001    | Electro World (National Chain) | C031    | Helsinki  | 1     | 1.38    | 12    | 1.49  |      | PR14       |             |
| 17 | P0017      | Pizza Oven        | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 12    | 1.49  |      | PR06       | high        |
| 18 | P0018      | Toaster Oven      | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 49    | 1.95  |      | PR14       |             |
| 19 | P0018      | Toaster Oven      | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 49    | 1.95  |      | PR14       |             |
| 20 | P0018      | Toaster Oven      | S0001    | Electro World (National Chain) | C031    | Helsinki  | 0     | 0       | 57    | 1.95  |      | PR14       |             |

salesproduct\_namesproduct\_hierarchycity\_namesstore\_namesstore\_cities

ReadyAccessibility: Investigate

100%

# Appendix: Regression analysis to estimate the relationship between sales\_date and sales



# Appendix: joining data sources on Tableau for in-dept analysis & visualisation

sales (1).csv+ (Multiple Connections)

Filters  
0 | Add

product\_hierarchy.csv

product\_names.csv

sales (1).csv

city\_names.csv

store\_cities.csv

store\_citie... — store\_nam...

Relationship

Related Fields

store\_cities.csv \* — \* store\_names.csv    Store Id (Store Cities.... = Store Id (Store Name...

Name store\_names.csv

These settings help Tableau optimize queries during analysis. The default settings are recommended, if you aren't sure what to choose. [Learn more](#)

Cardinality

Many — Many

Referential Integrity

Some records match — Some records match

store\_names.csv

Store Id (Store Names.Csv)

Store Name

S0005

MediaMarkt(National Chain)

S0036

Saturn

S0104

Euronics

S0068

FNAC

S0086

Darty



The End